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Radical reactions

Connections

➡ Building on

- Energy profile diagrams [ch12](#)
- Nucleophilic substitution [ch15](#)
- Conformational analysis [ch16](#)
- Elimination reactions [ch17](#)
- Conjugate addition [ch22](#)
- Regioselectivity [ch24](#)
- Retrosynthetic analysis [ch28](#)
- Diastereoselectivity [ch32 & ch33](#)
- Main group chemistry [ch27](#)

Arriving at

- Radicals are species with unpaired electrons
- Radical reactions follow different rules to those of ionic reactions
- Bond strength is very important
- Radicals can be formed with I, Br, Cl, Sn, and B
- Efficient radical reactions are chain reactions
- There are electrophilic and nucleophilic radicals
- Radicals favour conjugate addition
- Cyclization is easy with radical reactions

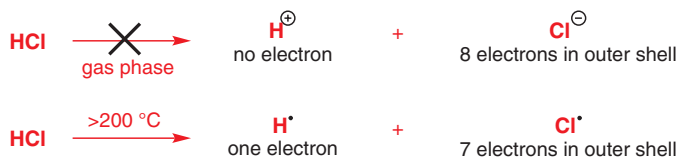
➡ Looking forward to

- Carbene chemistry [ch38](#)
- Determination of mechanism [ch39](#)
- Natural products [ch42](#)
- Polymerization [web](#)

Radicals contain unpaired electrons

You may remember that at the beginning of Chapter 8 we said that the cleavage of H–Cl into H^+ and Cl^- is possible in solution only because the ions that are formed are solvated: in the gas phase, the reaction is endothermic with $\Delta G = +1347 \text{ kJ mol}^{-1}$, a value so vast that even if the whole universe were made of gaseous HCl at 273 K, not a single molecule would be dissociated into H^+ and Cl^- ions. At temperatures above about 200 °C, however, HCl does begin to dissociate, but not into ions. Instead of the chlorine atom taking both bonding electrons with it, leaving a naked proton, the electron pair forming the H–Cl bond is shared out between the two atoms. ΔG for this reaction is a much more reasonable $+431 \text{ kJ mol}^{-1}$ and, at high temperatures (above about 200 °C, that is), HCl gas can be dissociated into H and Cl atoms.

■ The single, unpaired electron possessed by each atom is represented by a dot. The Cl atom, of course, has another three pairs of electrons that are not shown.

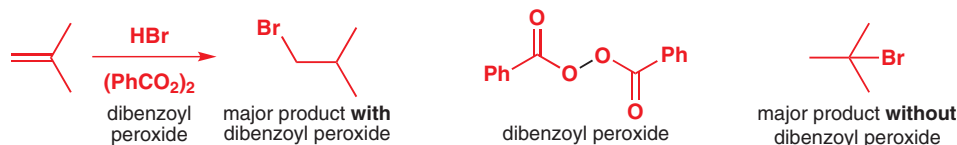


● Heterolysis and homolysis

- When bonds break and one atom gets both bonding electrons, the process is called **heterolysis**. The products of heterolysis are, of course, **ions**.
- When bonds break and the atoms get one bonding electron each, the process is called **homolysis**. The products of homolysis are **radicals**, which may be atoms or molecules, but must contain an unpaired electron.

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

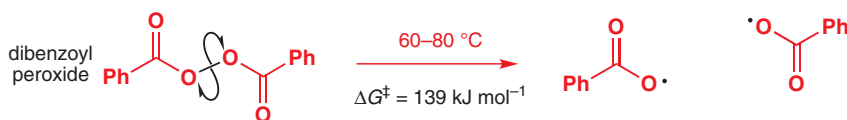
In Chapter 24 we introduced the fact that bromine radicals react regioselectively with alkenes. Let us remind you of one reaction you met then: radical addition to an alkene. The product is an alkyl bromide, and is a different alkyl bromide from the one formed when HBr adds to an alkene in an ionic manner.



What does the peroxide do to change the mechanism of the reaction? Peroxides undergo homolysis of the weak O—O bond extremely easily to form two radicals. We said that HCl in the gas phase undergoes homolysis in preference to heterolysis: other types of bond are even more susceptible to homolysis. You can see this for yourself by looking at this table of bond dissociation energies (ΔG for $\text{X}-\text{Y} \rightarrow \text{X}^\bullet + \text{Y}^\bullet$).

Bond X—Y	ΔG for $\text{X}-\text{Y} \rightarrow \text{X}^\bullet + \text{Y}^\bullet$, kJ mol ⁻¹	Bond X—Y	ΔG for $\text{X}-\text{Y} \rightarrow \text{X}^\bullet + \text{Y}^\bullet$, kJ mol ⁻¹
H—OH	498	CH ₃ —Br	293
H ₃ C—H	435	CH ₃ —I	234
H ₃ C—OH	383	Cl—Cl	243
H ₃ C—CH ₃	368	Br—Br	192
H—Cl	431	I—I	151
H—Br	366	HO—OH	213
H—I	298	MeO—OMe	151
CH ₃ —Cl	349		

Dialkyl peroxides (dimethyl peroxide is shown in the table) contain the very weak O—O bond. The radicals formed by homolytic cleavage of these bonds, stimulated by a little heat or light, initiate what we call a **radical chain reaction**, which results in the formation of the $\text{Br}^\bullet$ radicals, which add to the alkene's C=C double bond (see Chapter 24).



➔ Now would be a good time to revisit the section on radicals in Chapter 24 and to re-read pp. 571–574. Ionic additions to alkenes are covered in Chapter 19.

■ Try to get a feel for bond strengths: we shall refer to them a lot in this chapter as they're very important to radical reactions. Compare this with the situation for ionic reactions, in which the strengths of the bonds involved are often much less important than polar effects (see the example on p. 207).

Radicals form by homolysis of weak bonds

This is the most important way of making radicals: unpairing a pair of electrons by homolysis, making two new radicals. Temperatures of over 200 °C will homolyse most bonds; on the other hand, some weak bonds will undergo homolysis at temperatures little above room temperature. Light is a possible energy source for the homolysis of bonds too. Red light has associated with it 167 kJ mol⁻¹; blue light has about 293 kJ mol⁻¹. Ultraviolet (200 nm), with an associated energy of 586 kJ mol⁻¹, will decompose many organic compounds (including the DNA in skin cells: sunbathers beware!).

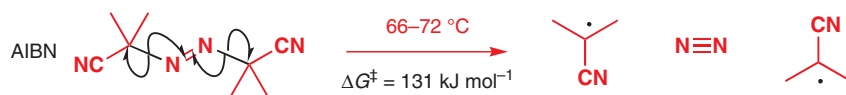
There are a number of compounds whose homolysis is particularly important to chemists, and the most important ones are discussed in turn below. They all have weak σ bonds, and generate radicals that can be put to some chemical use. The halogens are quite readily homolysed by light, as you can see from the bond strengths in the table above, a fact that drives the radical halogenation reactions that we shall discuss later.

As you saw in Chapter 24, dibenzoyl peroxide is an important compound because it can act as another initiator of radical reactions. It undergoes homolysis simply on heating.

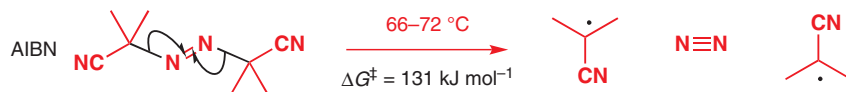
It is not sufficient for light to be energetic enough to promote homolysis; the molecule must have a mechanism for absorbing that energy, and the energy must end up concentrated in the vibrational mode that leads to bond breakage. We shall not consider these points further: if you are interested, you will find detailed explanations in specialized books on photochemistry.

Another compound that is often used in synthetic reactions for the same reason (although it reacts with a different set of compounds) is AIBN (azobisisobutyronitrile).

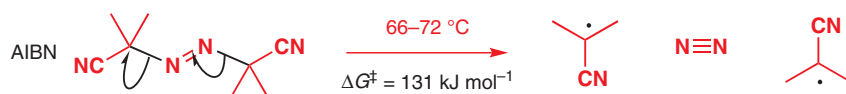
■ $\Delta G^\ddagger$ is the activation energy for the reaction.



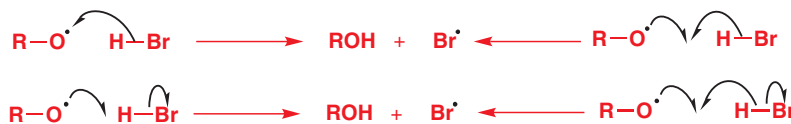
This decomposition mechanism accounts for the separate movements of all the electrons, but we can also draw the mechanism in a slightly different way: we show two radical ('fish hook') arrows forming the molecule of nitrogen but only one arrow to break each of the C–N bonds. It can be assumed that the electrons 'left behind' form radicals as well.



Another way of cutting back on the number of arrows without losing precision in the mechanism is to draw one arrow for each step all in the same (which can be either) direction. The first mechanism has the advantage of complete clarity; the other two make for neater diagrams. Choose which suits you best.



The important thing is to use the right type of arrow and to make it clear whether you are moving one, and not two, electrons. A simpler example is the abstraction of a hydrogen atom by an oxygen-centred radical: any of the mechanisms below is fine.



Radicals form by abstraction

Notice that we didn't put HBr on the list of molecules that form radicals by homolysis: relative to the weak bonds we have been talking about, the H–Br bond is quite strong (just about as strong as a C–C bond). We described in Chapter 24 how oxygen radicals abstract hydrogen atoms from HBr. You might now like to compare this mechanism with similar ionic reactions.



Hydrogen abstraction is the removal of a hydrogen atom **with its one electron**. It is not the removal of a proton: that would be the removal of a hydrogen atom **with no electrons**, which happens in ionic reactions.

The ability of radicals to propagate by abstraction is a key feature of radical chain reactions, which we shall come to later. There is an important difference between homolysis and abstraction as a way of making radicals: homolysis is a reaction of a spin-paired molecule that produces *two* radicals; abstraction is a reaction of a radical with a spin-paired molecule that produces *one* new radical and a new spin-paired molecule.

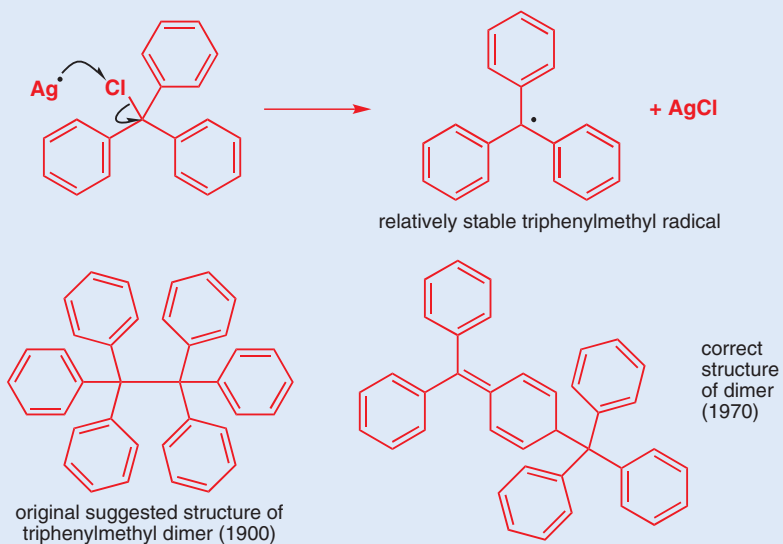
■ We use the term 'spin-paired molecule' to mean a 'normal' molecule, in which all the electrons are paired, in contrast with a radical, which has an unpaired electron.

As the comparison above shows, radical abstractions are in fact substitution reactions (at H in this case). However, radical substitutions differ considerably from S_N1 or S_N2 reactions: importantly, *radical substitutions almost never occur at carbon atoms*. We shall come back to radical substitutions, or abstractions (depending on whether you take the point of view of the H atom or the Br atom), and explain why this should be, later in the chapter.

First radical detected

The very first radical to be detected, the triphenylmethyl radical, was made in 1900 by abstraction of $\text{Cl}^\bullet$ from Ph_3CCl by Ag metal. Many metal atoms such as $\text{Ag}^\bullet$ and $\text{Li}^\bullet$ have single unpaired electrons.

This radical is relatively stable (we shall see why shortly), but reacts with itself reversibly in solution. The product of the dimerization of triphenylmethyl was for 70 years believed to be hexaphenyl ethane but, in 1970, NMR showed that it was, in fact, an unsymmetrical dimer.



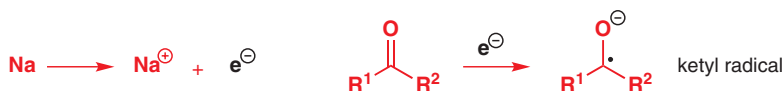
Radicals form by addition

The key step in the radical addition of HBr to an alkyne in Chapter 24 was the formation of a radical by radical addition. The $\text{Br}^\bullet$ radical (which, you will remember, was formed by abstraction of $\text{H}^\bullet$ from HBr by $\text{RO}^\bullet$) adds to the alkene to give a new, carbon-centred radical. This is the radical addition mechanism:



Just as charge must be conserved through a chemical reaction, so must the spin of the electrons involved. If a reactant carries an unpaired electron, then so must a product. Addition of a radical to a spin-paired molecule always generates a new radical. Radical addition is therefore a second type of radical-forming reaction.

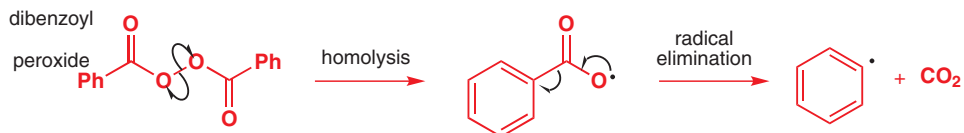
The simplest radical addition reactions occur when a single electron is added to a spin-paired molecule. This process is a reduction. You have already met some examples of single-electron reductions: Birch reductions (Chapter 23) use the single electron formed when a group I metal (sodium, usually) is dissolved in liquid ammonia to reduce organic compounds. Group I metals are common sources of single electrons: by giving up their odd s electron they form a stable M^+ ion. They will donate this electron to several classes of molecules, for example ketones can react with sodium to form ketyl radicals.



➡ We shall discuss ketyl radicals and their reactions on p. 980.

Radicals form by elimination

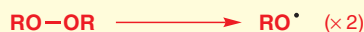
A fourth class of radical-forming reaction is **elimination**. For an example, we can go back to dibenzoyl peroxide, the unstable compound we considered earlier in the chapter. The radicals formed from dibenzoyl peroxide by homolysis are themselves unstable and each can break down by cleavage of a C–C bond, generating CO₂ and a phenyl radical. This is a radical elimination reaction, and is the reverse of a radical addition reaction.



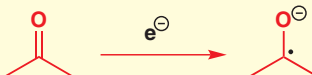
• To summarize methods of radical formation

Radicals form from spin-paired molecules by:

- homolysis of weak σ bonds, e.g.



- electron transfer, that is, reduction (addition of an electron), e.g.



Radicals form from other radicals by:

- substitution (abstraction)



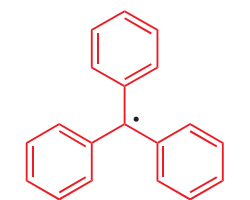
- addition



- elimination (homolysis)



■ *Electron célibataire* is the French term for these bachelor electrons searching earnestly for a partner.



triphenylmethyl radical—
stable in solution
in equilibrium with its dimer

Most radicals are extremely reactive...

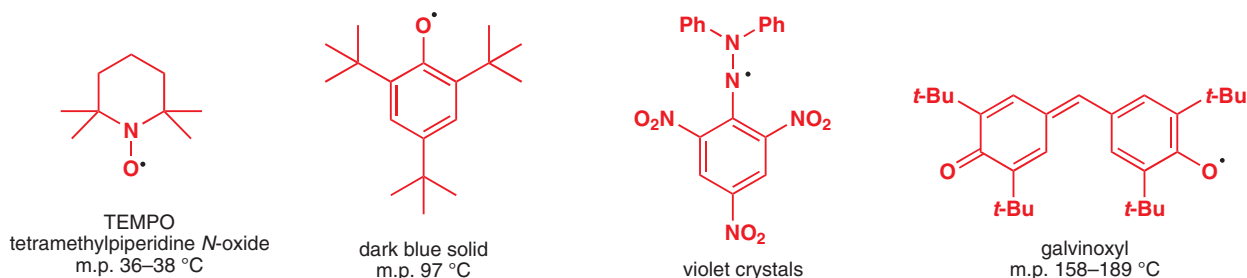
Unpaired electrons are desperate to be paired up again. This means that radicals usually have a very short lifetime; they don't survive long before undergoing a chemical reaction. Chemists are more interested in radicals that are reactive because they can be persuaded to do interesting and useful things. However, before we look at their reactions, we shall consider some radicals that are *unreactive* so that we can analyse the factors that contribute to radical reactivity.

... but a few radicals are very unreactive

Whilst simple alkyl radicals are extremely short-lived, some other radicals survive almost indefinitely. Such radicals are known as *persistent radicals*. We mentioned the triphenylmethyl

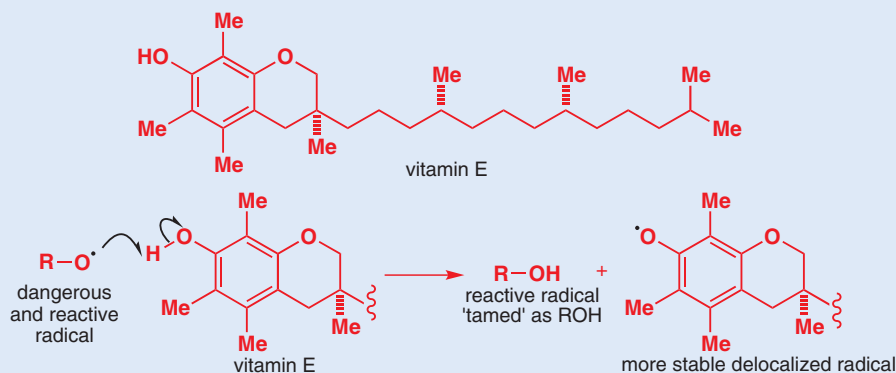
radical on p. 973: this yellow substance exists in solution in equilibrium with its dimer, but it is persistent enough to account for 2–10% of the equilibrium mixture.

Persistent radicals with the single electron carried by an oxygen or a nitrogen atom are also known: these four radicals can all be handled as stable compounds. The first, known as TEMPO, is a commercial product and can even be sublimed.



Vitamin E tames radicals

Many of the molecules that make up the structure of human tissue are susceptible to homolysis in intense light, and the body makes use of sophisticated chemistry to protect itself from the action of the reactive radical products. Vitamin E plays an important role in the 'taming' of these radicals: abstraction of H from the phenolic hydroxyl group produces a relatively stable radical that does no further damage.



There are two reasons why some radicals are more persistent than others: (1) steric hindrance and (2) electronic stabilization. In the four extreme cases above, their exceptional stability is conferred by a mixture of these two effects. Before we can analyse the stability of other radicals, however, we need to look at what is known about the shape and electronic structure of radicals.

How to analyse the structure of radicals: electron spin resonance

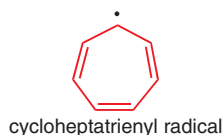
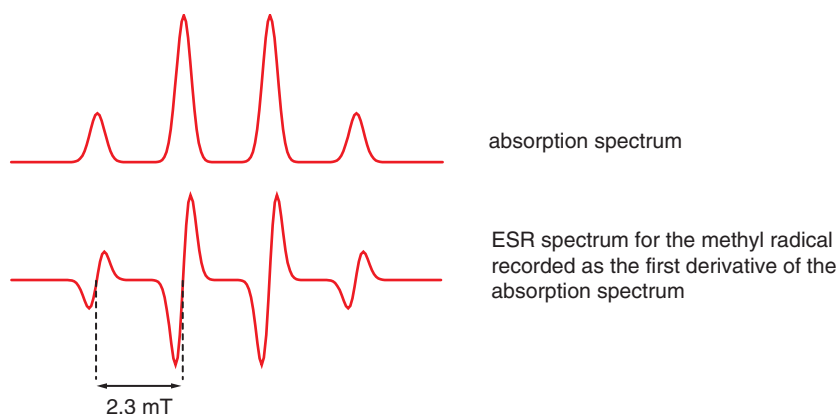
For the last few pages we have been discussing the species we call radicals without offering any evidence that they actually exist. Well, there is evidence, and it comes from a spectroscopic technique known as electron spin resonance, or ESR (also known as EPR, electron paramagnetic resonance). ESR not only confirms that radicals do exist, but it can also tell us quite a lot about their structure.

Unpaired electrons, like the nuclei of certain atoms, have a magnetic moment associated with them. Proton NMR probes the environment of hydrogen atoms by examining the energy difference between the two possible orientations of their magnetic moments in a magnetic field; ESR works in a similar way for unpaired electrons. The magnetic moment of an electron is much bigger than that of a proton, so the difference in energy between the possible quantum

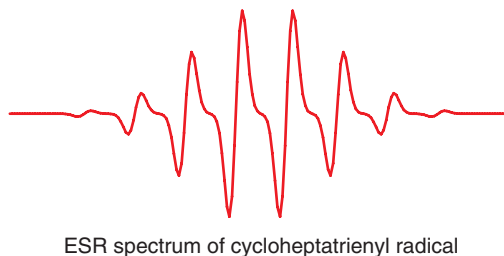
states in an electron field is also much bigger. This means that the magnets used in ESR spectrometers can be weaker than those in NMR spectrometers, usually about 0.3 tesla; even at this low field strength, the resonant frequency of an electron is about 9000 MHz (for comparison, the resonant frequency of a proton at 9.5 tesla is 400 MHz; in other words, a 400 MHz NMR machine has a magnetic field strength of 9.5 tesla).

But there are strong similarities between the techniques. ESR shows us, for example, that unpaired electrons couple with protons in the radical. The spectrum below is that of the methyl radical, $\text{CH}_3^\bullet$. The 1:3:3:1 quartet pattern is just what you would expect for coupling to three equivalent protons; coupling in ESR is measured in millitesla (or gauss; 1 gauss = 0.1 mT), and for the methyl radical the coupling constant (called a_{H}) is 2.3 mT.

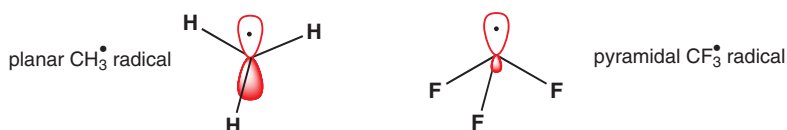
■ Notice that, for historical reasons, ESR spectra are recorded in a different way from NMR spectra: the diagram shows the first derivative of the absorption spectrum (the sort of spectrum you would get from a proton NMR machine).



ESR hyperfine splittings (as the coupling patterns are known) can give quite a lot of information about a radical. For example, here is the hyperfine splitting pattern of the cycloheptatrienyl radical. The electron evidently sees all seven protons around the ring as equivalent, and must therefore be fully delocalized. A localized radical would see several different types of proton, resulting in a much more complex splitting pattern.

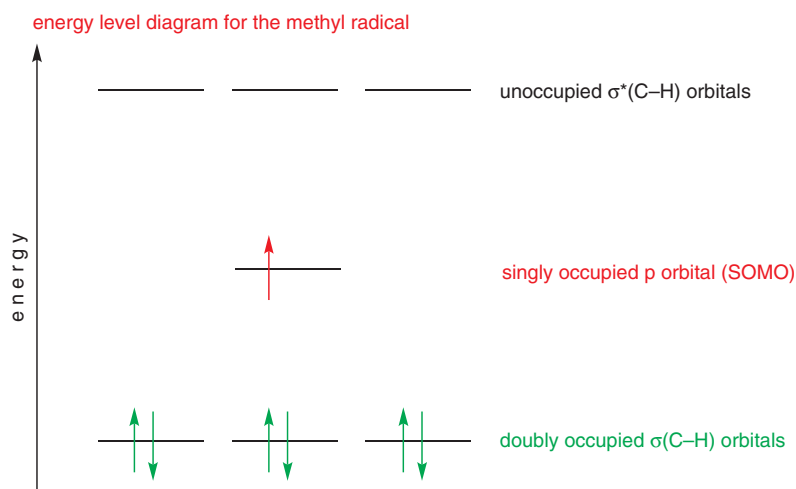


Even the relatively simple spectrum of the methyl radical tells us quite a lot about its structure. For example, the size of the coupling constant a_{H} indicates that the methyl radical is planar; the trifluoromethyl radical is, on the other hand, pyramidal. The oxygenated radicals $^\bullet\text{CH}_2\text{OH}$ and $^\bullet\text{CMe}_2\text{OH}$ lie somewhere in between. The calculations that show this lie outside the scope of this book.



Radicals have singly occupied molecular orbitals

ESR tells us that the methyl radical is planar: the carbon atom must therefore be sp^2 hybridized, with the unpaired electron in a p orbital. We can represent this in an energy level diagram.

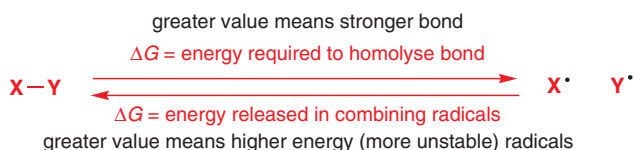


In Chapter 4 we talked about the HOMO (highest occupied molecular orbital) and LUMO (lowest unoccupied molecular orbital) of organic molecules. CH_3 (like all radicals) has an orbital containing one electron, which we call a **singly occupied molecular orbital (SOMO)**.

As with all molecules, it is the energy of the electrons in the molecular orbitals of the radical that dictate its stability. Any interaction that can decrease the energy levels of the filled molecular orbitals increases the stability of the radical (in other words, decreases its reactivity). Before we use this energy level diagram of the methyl radical to explain the stability of radicals, we need to look at some experimental data that allow us to judge just how stable different radicals are.

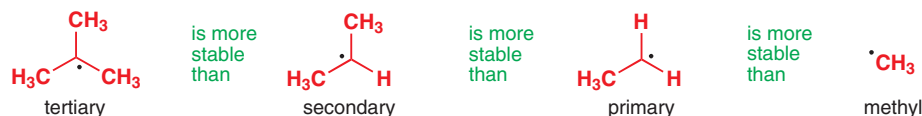
Radical stability

On p. 971 we used bond strength as a guide to the likelihood that bonds will be homolysed by heat or light. Since bond energies give us an idea of the ease with which radicals can form, they can also give us an idea of the stability of those radicals once they have formed.



This is particularly true if we compare the strengths of bonds between the same atoms, for example carbon and hydrogen, in different molecules; this table does this.

A few simple trends are apparent. For example, C–H bonds decrease in strength in R–H when R goes from primary to secondary to tertiary. Tertiary alkyl radicals are therefore the most stable; methyl radicals the least stable.



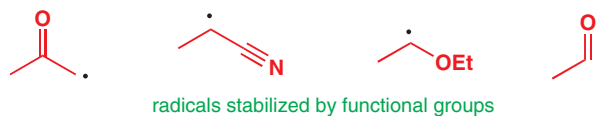
C–H bonds next to conjugating groups such as allyl or benzyl are particularly weak, so allyl and benzyl radicals are more stable. But C–H bonds to alkynyl, alkenyl, or aryl groups are strong. We saw the effects of this in Chapter 24.

C–H bond	Dissociation energy, kJ mol^{-1}
$\text{CH}_3\text{—H}$ (methyl)	439
$\text{MeCH}_2\text{—H}$ (primary)	423
$\text{Me}_2\text{CH—H}$ (secondary)	410
$\text{Me}_3\text{C—H}$ (tertiary)	397
$\text{HC}\equiv\text{C—H}$ (alkynyl)	544
$\text{H}_2\text{C}=\text{CH—H}$ (vinyl)	431
Ph—H (phenyl)	464
$\text{H}_2\text{C}=\text{CH}_2\text{CH}_2\text{—H}$ (allyl)	364
$\text{PhCH}_2\text{—H}$ (benzyl)	372
RC(=O)—H (acyl)	364
EtOCHMe—H	385
$\text{N}\equiv\text{CCH}_2\text{—H}$	360
$\text{MeCOCH}_2\text{—H}$	385

■ The absolute values in this table were determined in the gas phase, but the relative stabilities of the different radicals mirror their relative stabilities in solution.



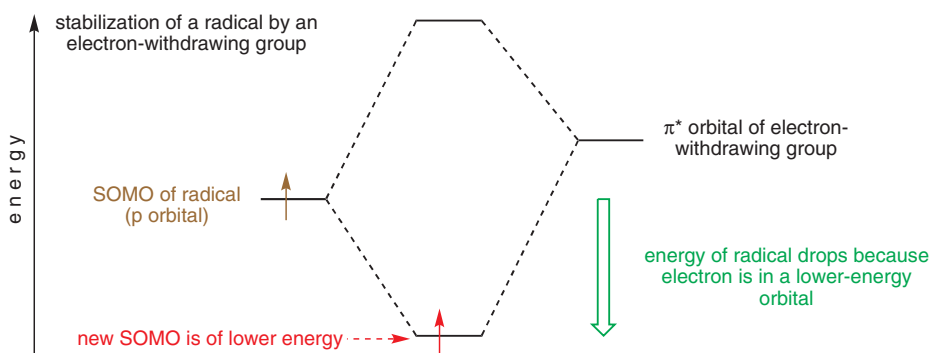
Adjacent functional groups appear to weaken C–H bonds: radicals next to carbonyl, nitrile, or ether functional groups, or centred on a carbonyl carbon atom, are more stable than even tertiary alkyl radicals.



Whether the functional group is electron withdrawing or electron donating is clearly irrelevant here: both types seem to stabilize radicals. We can explain all of this if we look at how the different groups next to the radical centre interact electronically with the radical.

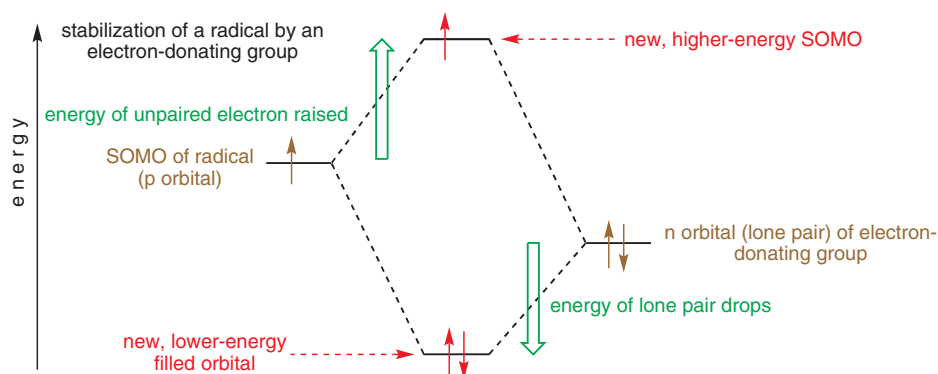
Radicals are stabilized by conjugating, electron-withdrawing, and electron-donating groups

Let's consider first what happens when a radical centre finds itself next to an electron-withdrawing group. Groups like C=O and C≡N are electron withdrawing because they have a low-lying empty π^* orbital. By overlapping with the (usually p) orbital containing the radical (the SOMO), two new molecular orbitals are generated. One electron (the one in the old SOMO) is available to fill the two new orbitals. It enters the new SOMO, which is of lower energy than the old one, and the radical experiences stabilization because this electron drops in energy.



Radicals that are stabilized by an electron-withdrawing group and an electron-donating group at the same time are sometimes known as **captodative radicals**.

We can analyse what happens with electron-rich groups, such as RO groups, in a similar way. Ether oxygen atoms have relatively high-energy filled n orbitals, their lone pairs. Interacting this with the SOMO again gives two new molecular orbitals. Three electrons are available to fill them. The SOMO is now higher in energy than it was to start with, but the lone pair is lower. Because two electrons have dropped in energy and only one has risen, there is an overall stabilization of the system, even though the new SOMO is of higher energy than the old one. We shall see later what effect the energy of the SOMO, rather than the overall energy of the radical, has on its reactivity.



In Chapter 15 you saw how the electrons in C–H σ bonds stabilize cations: they stabilize radicals in the same way, which is why tertiary radicals are more stable than primary ones. Conjugation, too, is effective at stabilizing radicals. We know from their ESR spectra (p. 976) that radicals next to double bonds are delocalized; that they are more stable is evident from the bond dissociation energies of allylic and benzylic C–H bonds.

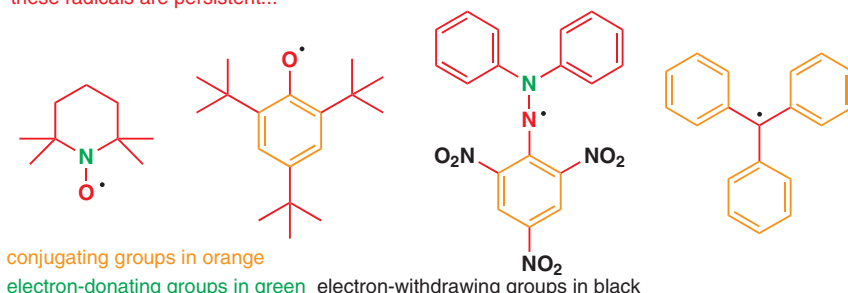
● **Anything that would stabilize an anion or a cation will stabilize a radical:**

- electron-withdrawing groups
- electron-donating groups (including alkyl groups with C–H σ bonds)
- conjugating groups.

Steric hindrance makes radicals less reactive

On p. 975 we showed you some radicals that are remarkably stable (persistent): some can even be isolated and purified. You should now be able to see at least part of the reason for their exceptional stability: two of them have adjacent powerful electron-donating groups, one has a powerful electron-withdrawing group as well, and three of the four are conjugated.

these radicals are persistent...

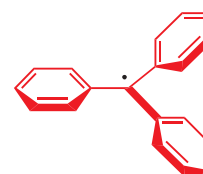


...while these radicals are reactive



But electronic factors alone are not sufficient to explain the exceptional stability of all four radicals, since the two radicals on the right receive just about the same electronic stabilization as the first two above, but are much more reactive.

In fact, the stability of the triphenylmethyl radical we know to be due mainly to steric, rather than electronic, factors. X-ray crystallography shows that the three phenyl rings in this compound are not coplanar but are twisted out of a plane by about 30°, like a propeller. This means that the delocalization in this radical is less than ideal (we know that there is some delocalization from the ESR spectrum) and, in fact, it is little more delocalized than the diphenylmethyl or even the benzyl radical, even though it is much more stable than either. This must be because the central carbon, which bears most of the radical character, is sterically shielded by the twisted phenyl groups, making it very hard for the molecule to react.



■ When it does react, as you saw in the box on p. 973, it does so through one of the less hindered *para* positions.

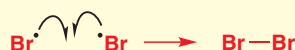
The rest of this chapter is devoted to the reactions of radicals, and you will see that the two effects we have talked about—electronic stabilization and steric hindrance—are key factors that control these reactions.

How do radicals react?

A reactive radical has a choice: it can either find another radical and combine to form a spin-paired molecule (or more than one spin-paired molecule), or it can react with a spin-paired molecule to form a new radical. Both are possible, and we shall see examples of each. A third alternative is for a radical to decompose in a unimolecular reaction, giving rise to a new radical and a spin-paired molecule.

• Three possibilities:

- radical + radical $\rightarrow$ spin-paired molecule



- radical + spin-paired molecule $\rightarrow$ new radical + new spin-paired molecule



- radical $\rightarrow$ new radical + spin-paired molecule

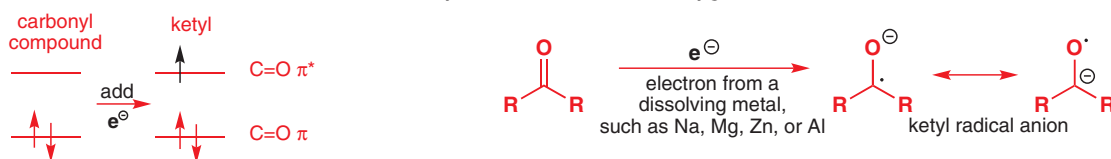


Radical–radical reactions

In view of the energy released when unpaired electrons pair up, you might expect this type of radical reaction to be more common than reaction with a spin-paired molecule, in which no net pairing of electrons takes place. Radical–radical reactions certainly do take place, but they are not the most important type of reaction involving radicals. We shall see why they are not as common as you might expect shortly, but first we can look at the few examples of radical–radical reactions which do work well.

The pinacol reaction is a radical dimerization

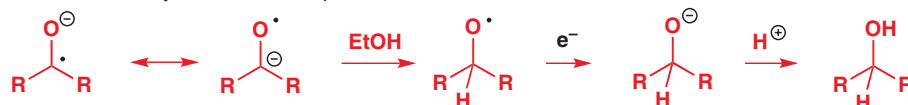
We outlined on p. 973 a way of making radicals by single electron transfer: effectively, the addition reaction of a single electron to a spin-paired molecule. The types of molecules that undergo this reaction are those with low-lying antibonding orbitals for the electron to go into, in particular aromatic systems and carbonyl compounds. The radical anion formed by addition of an electron to a ketone is known as a **ketyl**. The single electron is in the π^* orbital, so we can represent a ketyl with the radical on oxygen or on carbon and the anion on the other atom.



Ketyls behave in a manner that depends on the solvent that they are in. In protic solvents (ethanol, for example), the ketyl becomes protonated and then accepts a second electron from the metal (sodium is usually used in these cases). An alkoxide anion results, which, on addi-

tion of acid at the end of the reaction, gives an alcohol. Notice that this is a reaction using sodium metal in ethanol, and not sodium ethoxide, which is the basic product that forms once sodium has dissolved in ethanol. It is important that the sodium is *dissolving* as the reaction takes place, since only then are the free electrons available.

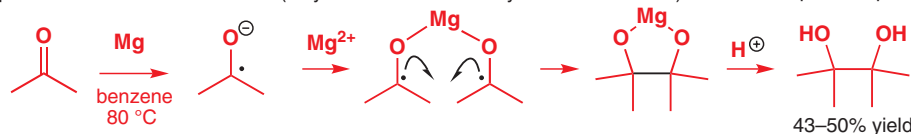
reaction of the ketyl radical anion in protic solvents



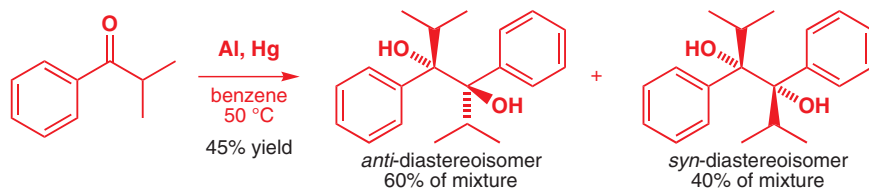
In aprotic solvents, such as benzene or ether, no protons are available so the concentration of ketyl radical builds up significantly and the ketyl radical anions start to dimerize. As well as being a radical–radical process, this dimerization process is an anion–anion reaction, so why doesn't electrostatic repulsion between the anions prevent them from approaching one another? The key to success is to use a metal such as magnesium or aluminium, which forms strong, covalent metal–oxygen bonds and can coordinate to more than one ketyl at once. Once two ketyls are coordinated to the same metal atom, they react rapidly.

pinacol dimerization of acetone (ketyl radical reaction in hydrocarbon solvent)

diol product 'pinacol'

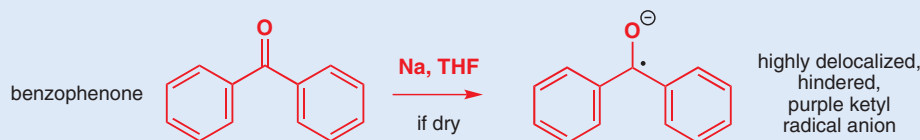


The example shows the dimerization of acetone to give a diol (2,3-dimethylbutane-2,3-diol) whose trivial name, pinacol, is used as a name for this type of reaction using any ketone. Sometimes pinacol reactions create new chiral centres: in this example, the two diastereoisomeric diols are formed in a 60:40 mixture. If you want to make a single diastereoisomer of a diol, a pinacol reaction is not a good choice!



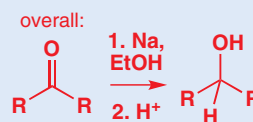
Benzophenone as an indicator in THF stills

As you should have gathered by now, THF is an important organic solvent in which many low-temperature, inert atmosphere reactions are conducted. It has a drawback, however: it is quite hygroscopic, and often the reactions for which it is used as a solvent must be kept absolutely free of water. It is therefore always distilled immediately before use from sodium metal, which reacts with any traces of water in the THF. However, it is necessary to have an indicator to show that the THF is dry and that the sodium has done its job. The indicator used is a ketone, benzophenone.



When the THF is dry, the distilling liquid containing the benzophenone becomes bright purple. This colour is due to the ketyl of benzophenone, the formation of which under these conditions should not surprise you. It should also come as no surprise that this ketyl, being stabilized by conjugation and quite hindered, is persistent (long-lived)—it does not undergo pinacol dimerization (as we explained above, you would not normally choose sodium to promote pinacols anyway). However, if water is present, the ketyl is rapidly quenched in the manner of the reduction described above to give the (colourless) alkoxide anion: only when all the water is consumed does the colour return.

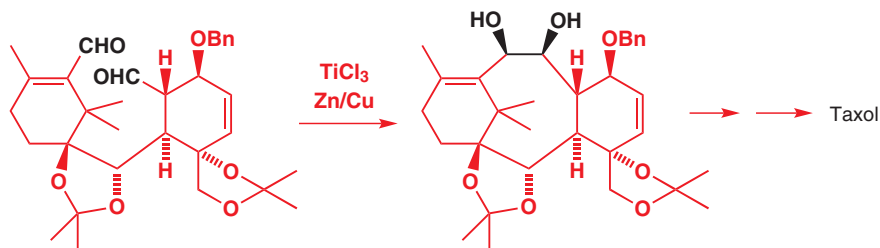
This reaction, known as the **Bouveault–Blanc reduction**, was used to reduce carbonyl compounds to alcohols, but now aluminium hydrides and borohydrides are usually more convenient. You met an example of the Bouveault–Blanc reduction in Chapter 32 (p. 832).



Interactive mechanism for pinacol reaction

■ You would be better off using one of the methods described in Chapter 33 on diastereoselectivity.

Pinacol reactions can be carried out intramolecularly, from compounds containing two carbonyl groups. In fact, the key step of one of the very first syntheses of the important anti-cancer compound Taxol was an intramolecular pinacol reaction using titanium as the source of electrons.

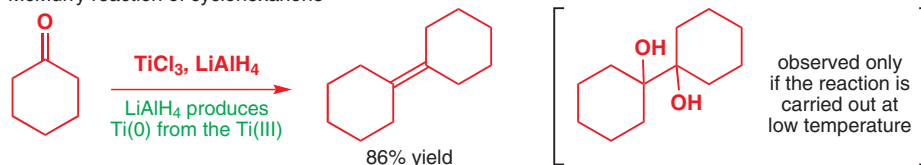


The titanium metal that is the source of electrons is produced during the reaction by reduction of TiCl_3 using a zinc–copper mixture. This reaction is, in fact, unusual because, as we shall see below, pinacol reactions using titanium do not normally stop at the diol, but give alkenes.

Titanium promotes the pinacol coupling and then deoxygenates the products: the McMurry reaction

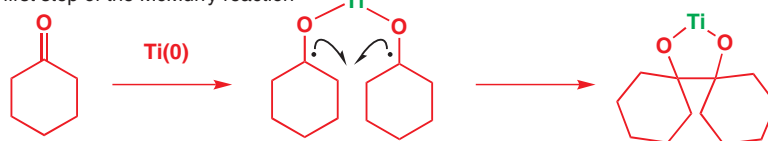
Titanium can be used as the metal source of electrons in the pinacol reaction and, provided the reaction is kept cold and not left for too long, diols can be isolated from the reaction, as in the example above. However, unlike magnesium or aluminium, titanium reacts further with these diol products to give alkenes in a reaction known as the McMurry reaction, after its inventor.

McMurry reaction of cyclohexanone

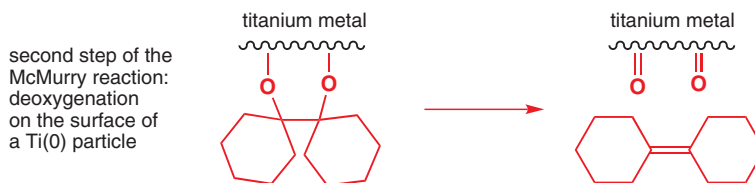


Notice that the titanium(0), which is the source of electrons in the reaction, is produced during the reaction by reacting a Ti(III) salt, usually TiCl_3 , with a reducing agent such as LiAlH_4 or Zn/Cu . The reaction does not work with, say, powdered titanium metal. The McMurry reaction is believed to be a two-stage process involving firstly a pinacol radical–radical coupling.

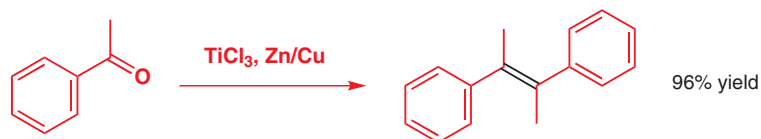
first step of the McMurry reaction



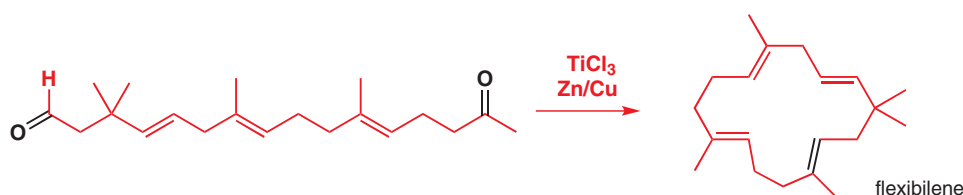
The Ti(0) then proceeds to deoxygenate the diol by a mechanism not fully understood, but thought to involve binding of the diol to the surface of the Ti(0) particles produced in the reduction of TiCl_3 .



We expect you to be mildly horrified by the inadequacy of the mechanism above. But, unfortunately, we can't do much better because no-one really knows quite what is happening. The McMurry reaction is very useful for making tetrasubstituted double bonds—there are few other really effective ways of doing this. However, the double bonds really need to be symmetrical (in other words, have the same substituents at each end) because McMurry reactions between two different ketones are rarely successful.

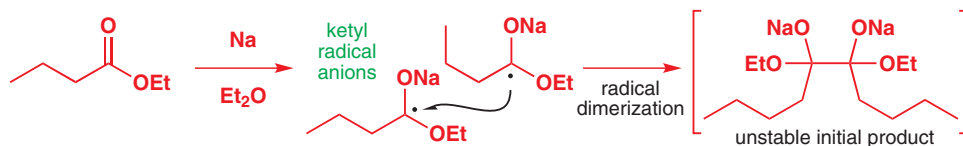


McMurry reactions also work very well intramolecularly, and turn out to be quite a good way of making cyclic alkenes, especially when the ring involved is medium or large (over about eight members). For example, the natural product flexibilene, with a 15-membered ring, can be made by cyclizing a 15-keto-aldehyde.

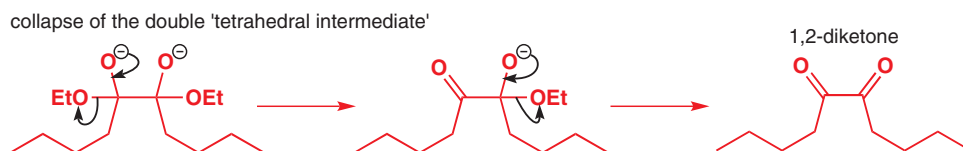


Esters undergo pinacol-type coupling: the acyloin reaction

You've seen examples of pinacol and McMurry reactions of ketones and aldehydes. What about esters? You would expect the ketyl radical anion to form from an ester in the same way, and then to undergo radical dimerization, and this is indeed what happens.

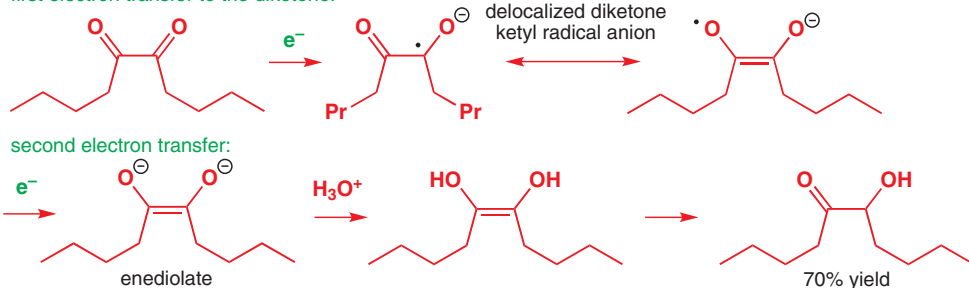


The product of the dimerization looks very much like a tetrahedral intermediate in a carbonyl addition–elimination reaction, and it collapses to give a 1,2-diketone.

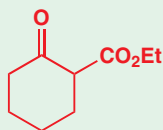


The diketone is, however, still reducible—in fact, 1,2-diketones are more reactive towards electrophiles and reducing agents than ketones because their π^* is lower in energy and straight away two electron transfers take place to form a molecule, which we could term an **enediolate**. On quenching the reaction with acid, this dianion is protonated twice to give the enol of an α -hydroxy-ketone, and it is this α -hydroxy-ketone that is the final product of the acyloin reaction. The yield in this example is a quite respectable 70%. However, in many other cases, this usefulness of the acyloin reaction is hampered by the formation of by-products that arise because of the reactivity of the enediolate dianion. It is, of course, quite nucleophilic, and is likely to be formed in the presence of the highly electrophilic diketone. It is also basic, and often catalyses a competing Claisen condensation of the esters being reduced.

first electron transfer to the diketone:

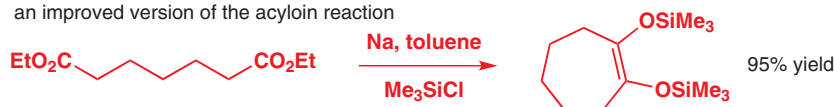


■ In the absence of the Me_3SiCl , the main product from this reaction becomes the cyclic ketoester below, which arises from base-catalysed Dieckmann cyclization (see Chapter 26) of the diester.

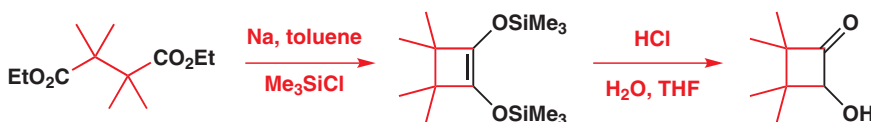


The solution to these problems is to add trimethylsilyl chloride to the reaction mixture. The silyl chloride silylates the enediolate as it is formed, and the product of the acyloin reaction becomes a bis-silyl ether.

an improved version of the acyloin reaction



These silyl ethers are rarely wanted as final products, and they can easily be hydrolysed to α -hydroxyketones with aqueous acid. This improved version makes four-membered rings efficiently.



It's not by accident that these two examples of the acyloin reaction show the formation of cyclic compounds. It is a particularly powerful method of making carbocyclic rings from four-membered upwards: the energy to be gained by pairing up the two electrons in the radical-radical reaction step more than compensates for the strain that may be generated in forming the ring.

The pinacol, McMurry, and acyloin reactions are exceptional

■ Think of radicals as smash-and-grab raiders. They pick the first shop that catches their eye, smash the window, and run off with a handful of cheap jewellery from the front of the display. Ions in solution are stealthy burglars. They scan all the houses on the street, choose the most vulnerable, and then carefully gain entry to the room that they know contains the priceless oil painting.

We've already said that this type of reaction, in which two radicals dimerize, is relatively uncommon. Most radicals are simply too reactive to react with one another! This may sound nonsensical, but the reason is simply that highly reactive species are unselective about what they react with. Although it might be energetically favourable for them to find another radical and dimerize, they are much more likely to collide with a solvent molecule, or a molecule of some other compound present in the mixture, than another radical. Reactive radicals are only ever present in solution in very low concentrations, so the chances of a radical-radical collision are very low. Radical attack on spin-paired molecules is much more common and, because the product of such reaction is also a radical, they give rise to the possibility of radical chain reactions.

Radical chain reactions

In looking at how radicals form, you've already seen examples of how radicals react. In fact, we've already dealt (if only very briefly) with every step of the sequence of reactions that makes up the mechanism of the radical reaction you met at the beginning of the chapter, and shown below.



Let's now consider each step in turn and in more detail.

1. The dialkyl peroxide is homolysed (by heat or light) to give two alkoxy radicals.



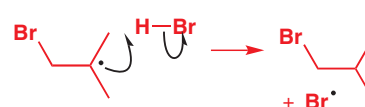
2. $\text{RO}^\bullet$ abstracts H from HBr (radical substitution) to give $\text{Br}^\bullet$.



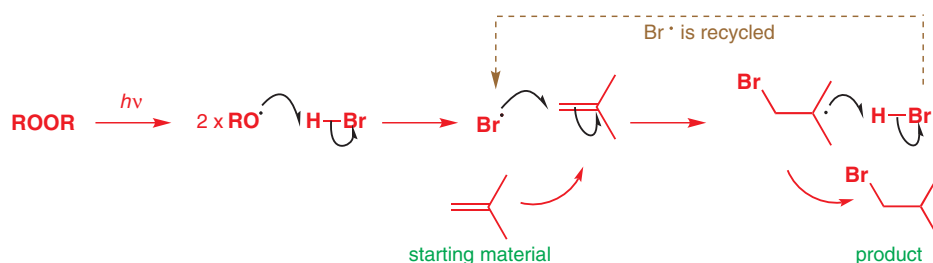
3. $\text{Br}^\bullet$ adds to isobutene to give a carbon-centred radical.



4. The carbon-centred radical abstracts a hydrogen atom from H-Br to form the final addition product and regenerate $\text{Br}^\bullet$, which can react with another molecule of alkene.



So the whole process is a cycle with the bromine radical regenerated in the last step, the one in which the product is formed.

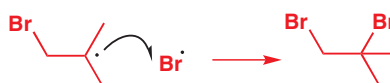


Interactive mechanism for radical addition of HBr to isobutene

In each step in the cycle a radical is consumed and a new radical is formed. This type of reaction is therefore known as a **radical chain reaction**, and the two steps that form the cyclic process that keeps the chain running are known as the **chain propagation steps**. Only one molecule of peroxide **initiator** is necessary for a large number of product molecules to be formed and, indeed, the peroxide needs to be added in only catalytic quantities (about 10 mol%) for this reaction to proceed in good yield.

Any less than 10 mol%, however, and the yield drops. The problem is that the chain reaction is not 100% efficient. Because the concentration of radicals in the reaction mixture is low, radical-radical reactions are rare, but nonetheless they happen often enough that more peroxide keeps being needed to start the chain off again.

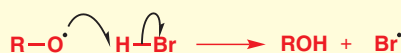
possible radical-radical chain termination steps



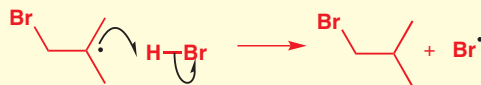
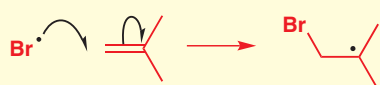
Reactions like this are known as **termination steps** and are actually an important part of any chain reaction; without termination steps the reaction would be uncontrollable.

● Radical chain reactions consist of:

• initiation steps

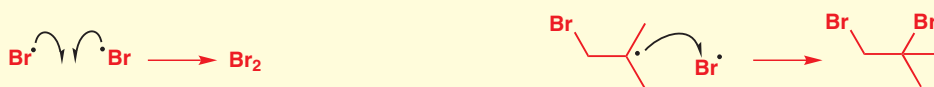


• propagation steps



Interactive mechanism for radical termination steps

• termination steps



■ We have already suggested two reasons why the $\text{Br}^\bullet$ radical adds to the alkene with this characteristic regioselectivity, giving a primary alkyl bromide when the polar addition of HBr to an alkene would give a tertiary alkyl bromide: (1) attack at the unsubstituted end of the alkene is less sterically hindered and (2) the tertiary radical thus formed is more stable than a primary radical. In fact, of all the hydrogen halides, only HBr will add to alkenes in this fashion: HCl and HI will undergo only polar addition to give the tertiary alkyl halide. Why? We need to be able to answer this type of question too.

Selectivity in radical chain reactions

In the radical–radical reactions we looked at earlier, there was never any question of what would react with what: only one type of radical was formed and the radicals dimerized in identical pairs. Look at the chain reaction above though—there are three types of radical present, $\text{Br}^\bullet$, $\text{BrCH}_2\text{Me}_2\text{CH}^\bullet$, and $\text{RO}^\bullet$, and they all react specifically with a chosen spin-paired partner: $\text{Br}^\bullet$ with the alkene, and $\text{BrCH}_2\text{Me}_2\text{CH}^\bullet$ and $\text{RO}^\bullet$ with HBr . We need to understand the factors that govern this chemoselectivity. In order to do so we shall look at another radical reaction with chemoselectivity and regioselectivity that is *measurable*.

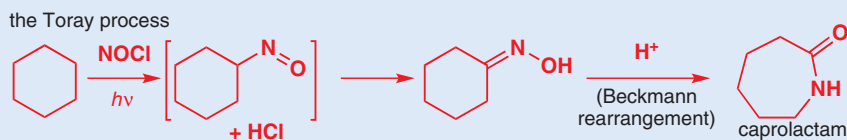
Chlorination of alkanes

Alkanes will react with chlorine radicals to give alkyl chlorides. For example, cyclohexane plus chlorine gas, in the presence of light, gives cyclohexyl chloride and hydrogen chloride.



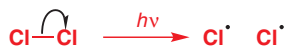
The Toray process

A variant of this reaction, known as the Toray process, is used on an industrial scale to produce caprolactam, a precursor to nylon. Instead of chlorine, nitrosyl chloride is used to form a nitroso compound that rapidly tautomerizes to an oxime. As you saw in Chapter 36, this oxime undergoes a Beckmann rearrangement under acid conditions to form caprolactam.

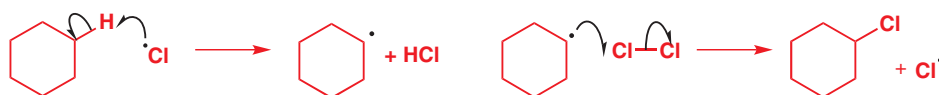


This type of reaction is important industrially since it is one of the few that allows compounds containing functional groups to be made from alkanes. As you might guess, since it needs light for initiation, the process is another example of a radical chain reaction. As with the radical addition of HBr to alkenes, we can identify initiation, propagation, and termination steps in the mechanism.

initiation



propagation



termination



Interactive mechanism for radical addition of Cl_2 to cyclohexane

In this case, the termination steps are much less important than in the last case we looked at, and typically the chain reaction can continue for 10^6 steps for each initiation

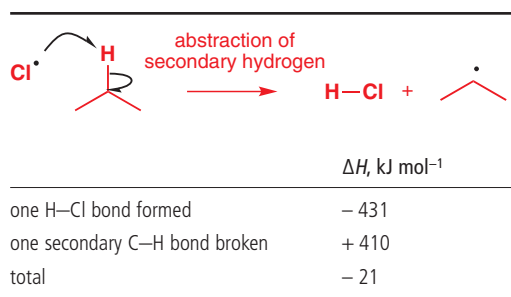
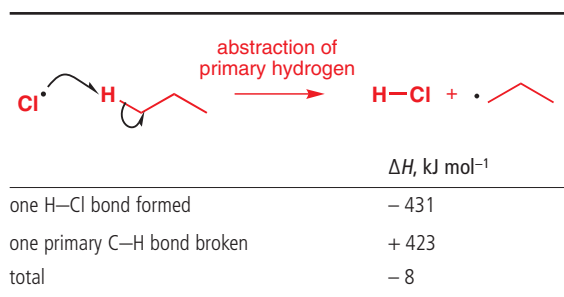
event (photolysis of chlorine). Be warned: reactions like this can be explosive in sunlight and are carried out in specialized facilities, not in the open laboratory.

When the chlorine radical abstracts a hydrogen atom from the cyclohexane, only one product can be formed because all 12 hydrogen atoms are equivalent. For other alkanes this may not be the case, and mixtures of alkyl chlorides can result. For example, propane is chlorinated to give a mixture of alkyl chlorides containing 45% 1-chloropropane and 55% 2-chloropropane, and isobutane is chlorinated to give 63% iso-butyl chloride and 37% *tert*-butyl chloride.

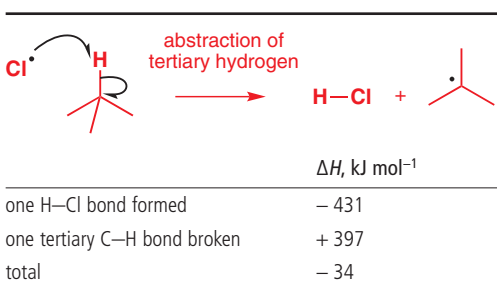
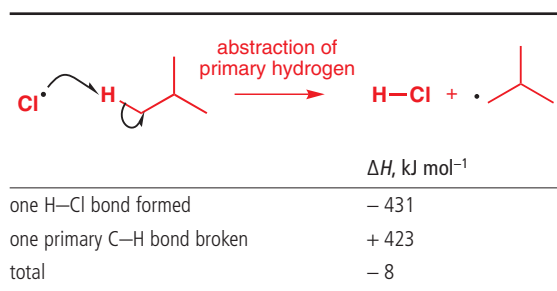


How can we explain the ratios of products that are formed? The key is to look at the relative stabilities of the radicals involved in the reaction and the strengths of the bonds that are formed and broken. First, the chlorination of propane. A chlorine radical, produced by photolysis, can abstract either a primary hydrogen atom, from the end of the molecule, or a secondary hydrogen atom, from the middle. For the two processes, we have these energy gains and losses:

➡ These bond energies were given in the tables on p. 971.



Abstraction of the secondary hydrogen atom is more exothermic than abstraction of the primary hydrogen atom for the related reasons that: (1) secondary C—H bonds are weaker than primary ones and (2) secondary radicals are more stable than primary ones. So, we get more 2-chloropropane than 1-chloropropane. But in this case, that isn't the only factor involved: remember that there are six primary hydrogen atoms and only two secondary ones, so the relative reactivity of the primary and secondary positions is even more different than the simple ratio of products from the reaction suggests. This statistical factor is more evident in the second example we gave above, the chlorination of isobutane. Now the choice is between formation of a tertiary radical and formation of a primary one.



Tertiary radical formation is more exothermic, yet more primary alkyl chloride is formed than tertiary alkyl chloride. However, once the 9:1 ratio of primary to tertiary hydrogen atoms is taken into account, the relative reactivities, as determined experimentally, turn out to be as shown in the table below.

ratio of products formed (tertiary:primary)	37:63
number of hydrogen atoms (tertiary:primary)	1:9
relative reactivity of each C—H bond (tertiary:primary)	$37/1.63/9 = 37:7 = \text{ca. } 5:1$

● Bond strength is important in radical reactions

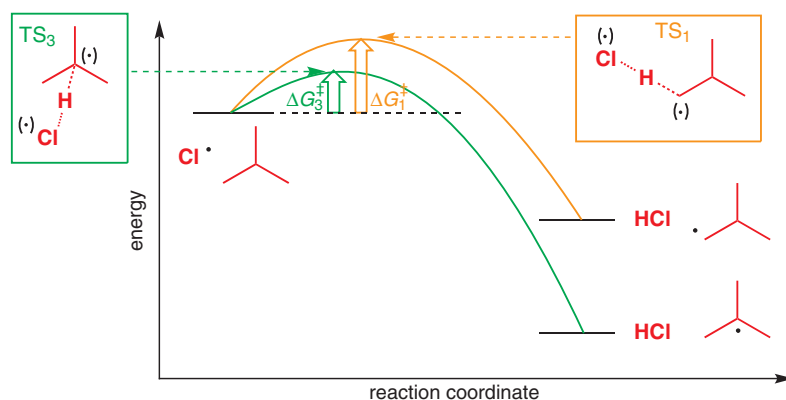
These reactions illustrate a key point about radical reactions—an important factor affecting selectivity is the strength of the bonds being formed and broken.

■ Bond strength is only a *guide* to selectivity in radical reactions. As we shall see shortly, it's not the only factor involved. Indeed, you've already seen *steric effects* in action when the $\text{Br}^\bullet$ radical is added to the less hindered end of the alkene in the first radical reaction of this chapter, and you will later see how *frontier orbital effects* can operate too.

■ We use the symbol $(\bullet)$ to mean a partial radical; a radical that is partially centred on this atom. The symbols $(-)$ and $(+)$ are used to mean a similar thing when a charge is shared by more than one atom.

■ Of course our calculations involving bond energies only gave us values for ΔH , not ΔG , which is what this diagram represents. However, we can assume that the $T\Delta S$ term in the relationship $\Delta G = \Delta H - T\Delta S$ is relatively insignificant.

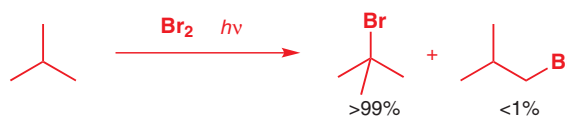
The rate of attack by $\text{Cl}^\bullet$ on a tertiary C—H bond, then, is about five times the rate of attack by $\text{Cl}^\bullet$ on a primary C—H bond. We said that this is because the formation of the tertiary radical is more exothermic than the formation of the primary radical. But the rate of a reaction depends not on ΔH for that reaction but on the activation energy of the reaction; in other words, the energy needed to reach the transition state for the reaction. But we can still use the stability of the product radicals as a guide to the stability of the transition state because the transition state must have significant radical character.



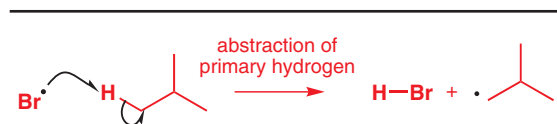
The energy diagram above illustrates this point. As the reactants ($\text{Cl}^\bullet$ plus isobutane) move towards the products, they pass through a transition state (TS_1 for formation of the primary radical, TS_3 for formation of the tertiary) in which the radical character of the $\text{Cl}^\bullet$ starting material is spread over both the Cl and the C centres. The greater stability of a tertiary radical compared with a primary one must be reflected to a lesser degree in these transition states: a radical shared between Cl and a tertiary centre will be more stable than a radical shared between Cl and a primary centre. The transition state TS_3 for the reaction at the tertiary C—H bond is therefore of lower energy than the transition state TS_1 for reaction at the primary C—H bond. In other words, the activation energy $\Delta G_3^\ddagger$ is smaller than $\Delta G_1^\ddagger$, so reaction at the tertiary C—H bond is faster.

Bromination of alkanes is more selective

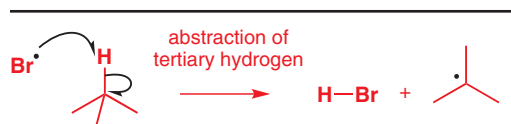
Bromine will also halogenate alkanes, and it does so much more selectively than chlorine. For example, the following reaction yields *tert*-butyl bromide with less than 1% of the primary isomer.



In this case, the first step of the radical chain reaction, the abstraction of H by $\text{Br}^\bullet$, is endothermic for both the primary and tertiary hydrogen atoms, but more so for primary radical formation, so the tertiary radical is preferred.



	$\Delta H, \text{ kJ mol}^{-1}$
one H—Br bond formed	− 366
one primary C—H bond broken	+ 423
total	+ 57



	$\Delta H, \text{ kJ mol}^{-1}$
one H—Br bond formed	− 366
one tertiary C—H bond broken	+ 397
total	+ 31

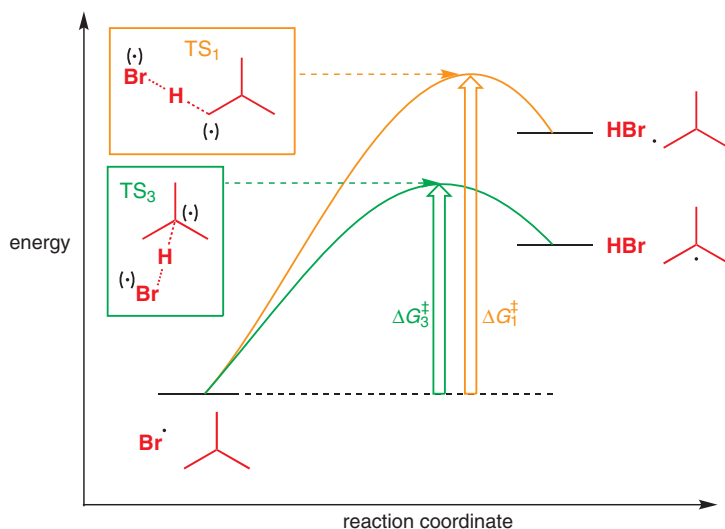
Of course, in both brominations and chlorinations the *overall* reaction is favourable because the second step—the halogenation of the alkyl radical—is significantly exothermic, by about 106 kJ mol^{-1} for chlorination and about 83 kJ mol^{-1} for bromination. The same is true for fluorination, but with fluorine this step is so exothermic that fluorination becomes dangerously explosive. Conversely, radical iodination is impossible because the final step is insufficiently exothermic to make up for the endothermic formation of an alkyl radical.

Why is bromination so much more selective than the chlorination of alkanes? This is a good example of the **Hammond postulate**, applied to real chemistry. Because the products of the first step of the bromination ($\text{R}^\bullet$ plus HBr) are higher in energy than the starting materials, the transition state must be similar in structure and energy to the radical being formed; the difference in energies of the primary and tertiary product radicals should therefore be markedly reflected in the different energies of the transition states TS_1 and TS_3 , and $\Delta G_1^\ddagger$ will be significantly larger than $\Delta G_3^\ddagger$. For the chlorination reaction, the products were just slightly lower in energy than the starting materials, so the transition states for the two possible reactions both resembled the starting materials rather more and the products rather less. These are the same for both tertiary and primary hydrogen abstractions, of course, so the difference in energy of the product radicals exerts a less pronounced effect on the difference in energy of the transition states.

second step of a halogenation:
always exothermic



The **Hammond postulate** gives information about the structure of transition states. It says that two states that inter-convert directly (are directly linked in a reaction profile diagram) and that are close in energy are also similar in structure. So a transition state will be most like the starting material, an intermediate, or the product if it is close in energy to one of these observable structures.



Allylic bromination

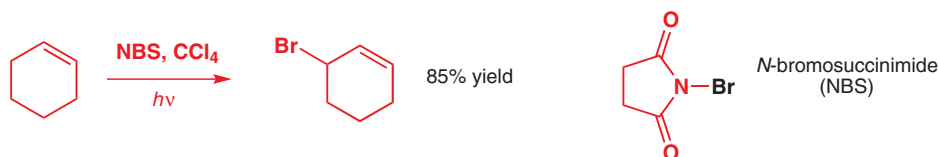
Because radical brominations are so selective, they can be used successfully in the laboratory to make alkyl bromides. There are relatively few ways of functionalizing an unfunctionalized centre, and radical allylic bromination is one of the most effective. We introduced this reaction in Chapter 24, where we contrasted the radical reactivity of Br_2 towards alkenes (leading to an allyl bromide by hydrogen abstraction) with its ionic reactivity (leading to addition of bromine across the alkene). We can now look in a little more detail at the selectivities involved.

➡ We introduced radical bromination on p. 571.

➡ The way that NBS does this is explained on p. 573.

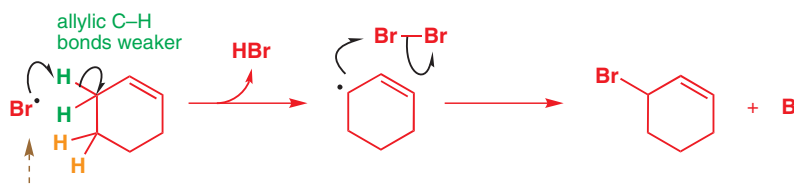
🖱 Interactive mechanism for allylic bromination

Here is a typical allylic bromination. NBS (*N*-bromosuccinimide) is used to form a small amount of Br_2 and to keep the Br_2 concentration low.

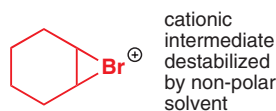
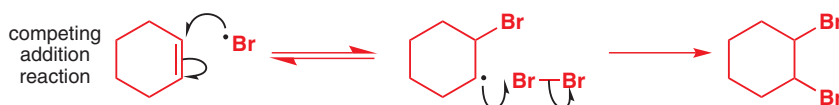


■ These figures were determined in the gas phase, and here our reactions are in solution. Nonetheless, because solvation effects are more or less the same for all radicals, we expect the order of the bond strengths to remain the same in both phases.

Photolysis of Br_2 initiates the reaction, which then propagates as shown below. The mechanism also illustrates the first aspect of selectivity: only a (green) allylic H atom is abstracted because an allylic C–H bond is considerably weaker than a secondary C–H bond (364 vs. 410 kJ mol^{-1} from the table on p. 977).

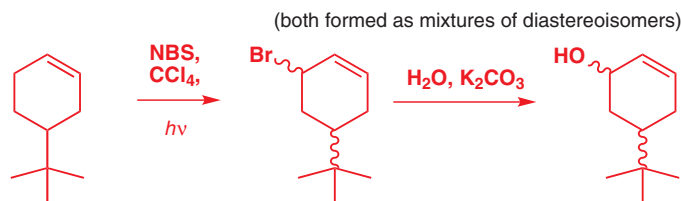


There is a problem with this reaction if bromine itself is used because an alternative radical addition reaction can compete with radical abstraction.



The first step of this competing addition reaction is reversible; the reaction is driven forward by the participation of a second molecule of bromine that traps the product alkyl radical. This side reaction can be prevented if the concentration of Br_2 in the reaction is kept very low, which is the role of NBS. The alternative competing polar addition of Br_2 to the alkene is likewise prevented with the low bromine concentration provided by NBS, although the non-polar solvent CCl_4 also disfavours the formation of the cationic bromonium ion intermediate.

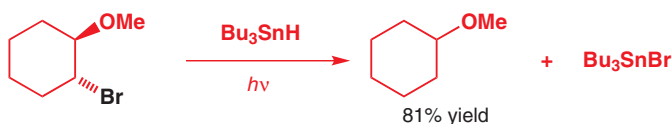
While radical halogenation of alkanes is used only rarely in the laboratory, radical allylic bromination of alkenes is a versatile and commonly used way of making allylic bromides. Nucleophilic substitution reactions can then be used to convert the bromide to other functional groups. For example, some chemists in Manchester needed to make the two diastereoisomers of 5-*tert*-butyl-cyclohex-2-en-1-ol to study their reactions with osmium tetroxide. *tert*-Butyl cyclohexene is readily available, so they used a radical allylic bromination to introduce the functional group in the allylic position, which they converted to a hydroxyl group using aqueous base. Steric effects also play a role here in the regioselectivity of the reaction: only the less hindered allylic hydrogen atoms further from the *t*-butyl group are removed.



Reversing the selectivity: radical substitution of Br by H

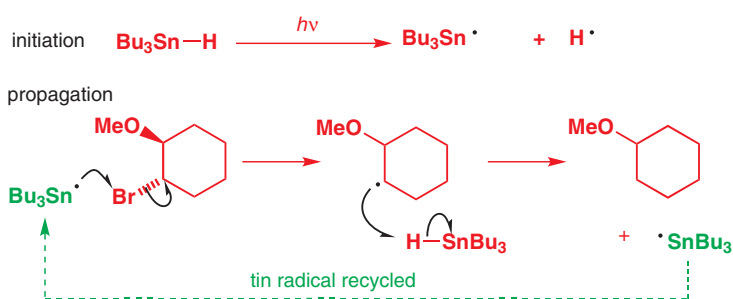
Radical substitution reactions can also be used to *remove* functional groups from molecules. A useful reagent for this (and, as you will see, for other radical reactions too) is tributyltin

hydride, Bu_3SnH . The Sn–H bond is weak and Bu_3SnH will react with alkyl halides to replace the halogen atom with H, producing Bu_3SnHal as a by-product.



➡ We discussed the removal of functional groups, and why you might want to do it, in Chapter 23.

Clearly, for this reaction to be energetically favourable, the new bonds formed (Sn–Br and C–H) must be stronger than the old bonds broken (Sn–H and C–halogen). Look at this table of average bond energies and you will see that this is indeed so. The use of a tin hydride is crucial to this reaction: Sn–H bonds are weaker than Sn–Br bonds, while, for carbon, C–H bonds are stronger. Bu_3SnH is therefore an effective source of $\text{Bu}_3\text{Sn}^\bullet$ radicals, and the $\text{Bu}_3\text{Sn}^\bullet$ radical will abstract halogens, particularly I or Br, but also Cl, from organic halides, breaking a weak C–halogen (C–Hal) bond and forming a strong Sn–Hal bond. The complete mechanism of the reaction reveals a chain reaction.

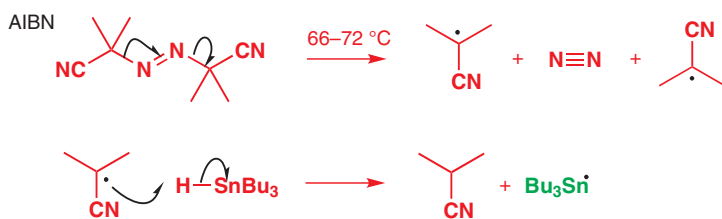


Bond	Representative bond energy, kJ mol^{-1}
C–Br	280
Sn–H	308
C–H	418
Sn–Br	552

🔗 Interactive mechanism for tin hydride reduction of alkyl halides

Homolysis of Bu_3SnH is promoted by the initiator AIBN

As you would imagine, the weakest C–Hal bonds are the easiest to cleave, so alkyl bromides are reduced more rapidly than alkyl chlorides, and alkyl fluorides are unreactive. With alkyl iodides and bromides, daylight can be sufficient to initiate the reaction, but with alkyl chlorides, and often with alkyl bromides as well, it is generally necessary to produce a higher concentration of $\text{Bu}_3\text{Sn}^\bullet$ radicals by adding an initiator to the reaction. The best choice is usually AIBN, which you met earlier in the chapter (p. 972). This compound undergoes thermal homolysis above 60°C to give nitrile-stabilized radicals that abstract the hydrogen atom from Bu_3SnH .

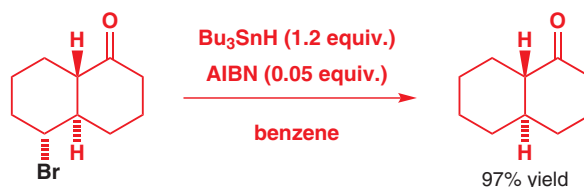


➡ We used peroxides as initiators of the addition of H–Br to alkenes on p. 971.

Why use AIBN as an initiator; why not a peroxide? Since we want to cleave only a weak Sn–H bond, we can get away with using a relatively unreactive nitrile-stabilized radical. Peroxides, on the other hand, generate $\text{RO}^\bullet$ radicals. These are highly reactive and will abstract hydrogen from almost any organic molecule, not just the weakly bonded hydrogen atom of Bu_3SnH , and this would lead to side reactions and lack of selectivity. AIBN is needed only in sufficient quantities to be an initiator of the reaction; it is the Bu_3SnH that provides the hydrogen atoms that end up in the product, so usually you need only 0.02 to 0.05 equivalents of AIBN and a slight excess (1.2 equivalents) of Bu_3SnH .

■ The bond energy of $\text{H–CH}_2\text{CN}$ is only 360 kJ mol^{-1} ; a tertiary C–H bond next to a CN group should be even weaker.

Bond energy of $\text{O–H} = 460 \text{ kJ mol}^{-1}$; few C–H bonds are stronger than 440 kJ mol^{-1} .



Carbon–carbon bond formation with radicals

You have now met these examples of radical chain reactions:

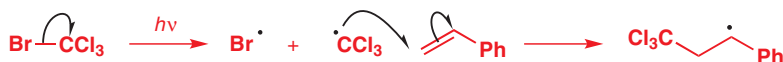
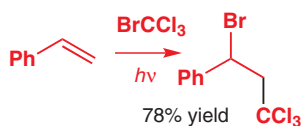
1. radical addition of halogens to double bonds
2. radical substitution of hydrogen by halogens, or of halogens by hydrogen.

You have seen how the selectivity of these reactions depends upon the bond strengths of the bond being formed or broken. Until about 1975, these reactions, with a few exceptions, were all that were expected of radicals. Since that date, however, the use of radicals in synthetic chemistry has increased tremendously, to the point where complex ring structures such as those of the natural product hirsutene or the steroids can be made from simple acyclic precursors in one radical-promoted step.

What has made this all possible is that chemists have learned how to understand the selectivity of radical reactions to such a degree that they can design starting materials and reagents to define precisely the bonds that will break and form during the reactions. We shall now go on to look at the most important consequence of this ability to control radical reactions: they can be used to make carbon–carbon bonds.

The radical reaction in the margin forms a new carbon–carbon bond. The mechanism is quite similar to that of the very first radical reaction we showed you, right at the beginning of the chapter. Now, with your additional appreciation of the role of bond strength in the selectivity of radical reactions, you should be able to understand why each step proceeds in the way that it does.

First the weakest bond, C–Br, is broken by the light being shone on to the reaction. Two radicals form, $\text{CCl}_3^\bullet$ and $\text{Br}^\bullet$, and it is the $\text{CCl}_3^\bullet$ that adds to the (less hindered) unsubstituted end of the alkene to produce a (more stable) secondary benzylic radical.



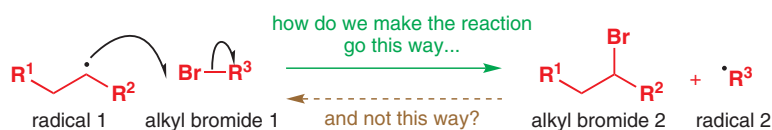
This radical abstracts a Br atom from the BrCCl_3 , breaking the (weakest) C–Br bond, forming the product, and regenerating $\text{CCl}_3^\bullet$, which adds to another molecule of alkene. Notice that the carbon-centred radical abstracts $\text{Br}^\bullet$ and not $\text{CCl}_3^\bullet$ from BrCCl_3 —to abstract $\text{CCl}_3^\bullet$ would require a radical substitution at carbon—remember, radicals want the easy pickings from the front of the display; they don't go nosing round the back to see if there's anything better to be had.



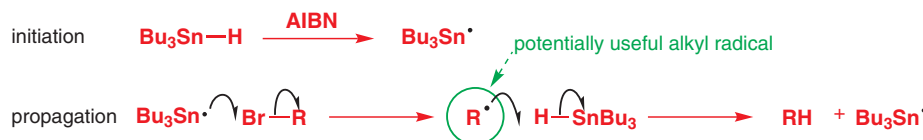
■ It is mainly this step that produces the $\text{CCl}_3^\bullet$ that undergoes addition to the alkene—the initial photolysis, of course, produces both $\text{Br}^\bullet$ and $\text{CCl}_3^\bullet$, either of which could add, but, once the radical chain has been initiated, only $\text{CCl}_3^\bullet$ is reproduced.

This reaction works quite well, giving 78% of the product, but it relies on the fact that the starting material, BrCCl_3 , has an unusually weak C–Br bond (the $\text{CCl}_3^\bullet$ radical is highly stabilized by those three chlorine atoms). You can't use most other alkyl bromides for a number of reasons, not least of them being that the product is also an alkyl bromide and, without the selectivity provided by the CCl_3 group, the result would be a mixture of polymers. The problem is that we want the product radical to abstract Br from the starting alkyl bromide to make

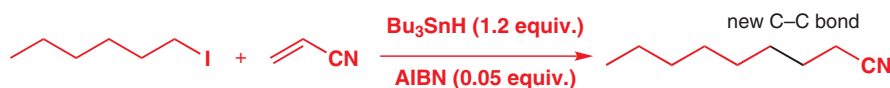
a new alkyl bromide and a new starting radical, and there is no energetic driving force behind this transformation.



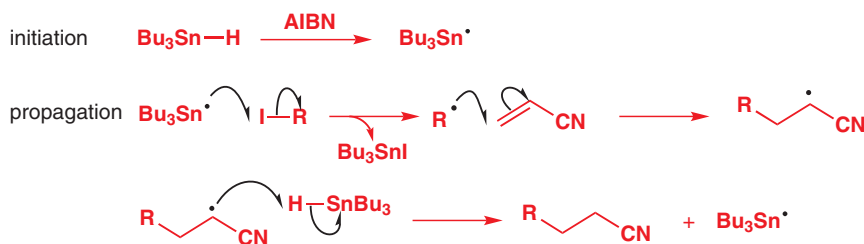
For a way of overcoming this problem, let's go back to the reaction we looked at a few pages ago, the dehalogenation of alkyl halides by Bu_3SnH . The mechanism involves formation of an alkyl (carbon-centred) radical by abstraction of Br by $\text{Bu}_3\text{Sn}^\bullet$. This alkyl radical then just abstracted $\text{H}^\bullet$ from Bu_3SnH .



Is it not possible to use this alkyl radical more constructively, and encourage it to react with another molecule (an alkene, say, as $\text{CCl}_3^\bullet$ did)? The answer is a qualified yes: look at this reaction:



We have added a carbon-centred radical to an alkene in a radical chain reaction! Here is the mechanism:



Something important has happened here: the product radical no longer has to abstract the halogen from the starting material, but instead has to abstract H from Bu_3SnH ; it is the $\text{Bu}_3\text{Sn}^\bullet$ thus formed that then regenerates the starting radical. The driving force is provided by formation of C-H at the expense of Sn-H and then Sn-Br at the expense of C-Br.

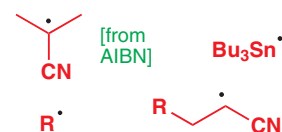
The use of tin hydrides increases the power of radical reactions in organic synthesis tremendously, and all of the steps in these radical chain processes have been studied in great detail because of the importance of the reactions. We won't dwell excessively on these details, but we need to go back and re-examine some points about this reaction because there are some further subtleties that you need to understand. Bear in mind that we have four radicals all in the reaction mixture at the same time. Yet each reacts with its chosen partner, forsaking all others.

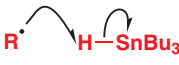
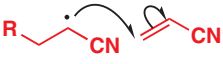
Let's take each type of radical in turn, and look at its selectivity. Clearly bond strength has something to do with this, but how do you explain the opposing selectivities of $\text{R}^\bullet$ and the nitrile-stabilized radicals? We shall see that the origins of the selectivities impose some restrictions on the type of starting material that can be used for these C-C bond-forming reactions.

 Interactive mechanism for radical addition of an alkyl group to acrylonitrile

➡ We explained on p. 991 how these same favourable thermodynamics drove the Bu_3SnH -promoted reduction of alkyl halides.

four radicals in the mixture:



Radical	Reacts like this	Does not react like this
$\text{Bu}_3\text{Sn}^\bullet$		
$\text{R}^\bullet$		
$\text{R}-\text{CH}_2-\text{CN}^\bullet$		
$\text{NC}-\text{C}(\text{Me})_2^\bullet$		

For the **tin radical**, $\text{Bu}_3\text{Sn}^\bullet$, there is a choice of reaction partners: we need it to abstract the halide from the starting material, but it could alternatively add to the alkene. The $\text{Sn}-\text{C}$ bond is relatively weak, so addition to the alkene becomes a significant reaction only if:

- there is a large excess of alkene present, and
- the starting alkyl halide is relatively unreactive. This means that only alkyl bromides and iodides can be used effectively to form carbon–carbon bonds; alkyl chlorides are just too unreactive.

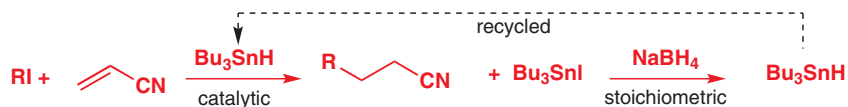
The contrasting reactivity of the **alkyl radical** $\text{R}^\bullet$ and the **nitrile-containing radicals** needs a little more analysis, and we will look at how both concentration and electronics affect their selectivities.

Concentration effects

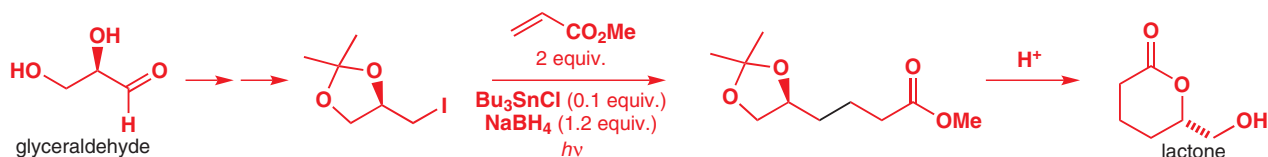
You know that $\text{R}^\bullet$ is perfectly capable of abstracting H from Bu_3SnH because that is what happened in the dehalogenation reaction on p. 991, but here a different reaction happens: addition to the alkene. In fact, the rate *constant* for reaction of $\text{R}^\bullet$ with Bu_3SnH is about the same as that for reaction with acrylonitrile ($\text{CH}_2=\text{CHCN}$), so the only way in which good yields can be obtained is by ensuring that the concentration of acrylonitrile is always at least 10 times that of the tin hydride. The difference in rates will then be sufficient to give 10 times as much addition to the alkene as reduction by the tin hydride. Too much acrylonitrile in the reaction mixture causes problems with side reactions, so a good way of achieving this is to add the tin hydride very slowly during the reaction—often a device known as a syringe pump is used for this. Of course, for complete reaction, a whole equivalent of hydride is necessary, but this can be added over a period of hours.

An elegant alternative is to use a technique conceptually similar to the use of NBS to provide a low concentration of Br_2 for radical allylic substitution. Instead of adding one equivalent of Bu_3SnH , a catalytic amount (usually 0.1–0.2 equivalents) of Bu_3SnCl is added at the beginning of the reaction, with 1 equivalent of NaBH_4 . NaBH_4 will reduce Bu_3SnHal to Bu_3SnH , so about 0.1 equivalent of Bu_3SnH is formed immediately. With each cycle of the chain reaction, a molecule of this Bu_3SnH is converted to Bu_3SnBr , which NaBH_4 can reduce back to Bu_3SnH . Only as much Bu_3SnH is produced as is needed because the rate of production is limited by the rate of reaction.

■ A useful alternative to NaBH_4 as a reducing agent, particularly when there are reactive carbonyl groups in the molecule, is NaCNBH_3 , which still reduces Bu_3SnHal but will not touch aldehydes or ketones (see Chapter 23).

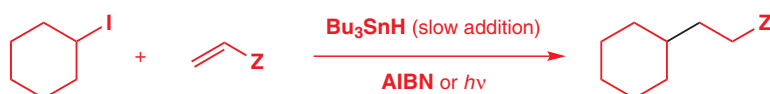


This method was used in the following example, in which an enantiomerically pure lactone, a useful synthetic building block, was made from naturally occurring glyceraldehyde.



Frontier orbital effects

The second key to success in making sure that the alkyl radical behaves well is to use a reactive radical trap. In fact, this is a major limitation of intermolecular radical carbon–carbon bond-forming reactions: for the trapping of alkyl radicals only electrophilic alkenes (attached to electron-withdrawing groups such as $-\text{CN}$, $-\text{CO}_2\text{Me}$, and $-\text{COMe}$) will do. This is a limitation, but nonetheless cyclohexyl iodide adds to all these alkenes with the yields shown and the rate of addition to most of these alkenes is 10^3 to 10^4 times that of addition to 1-hexene.



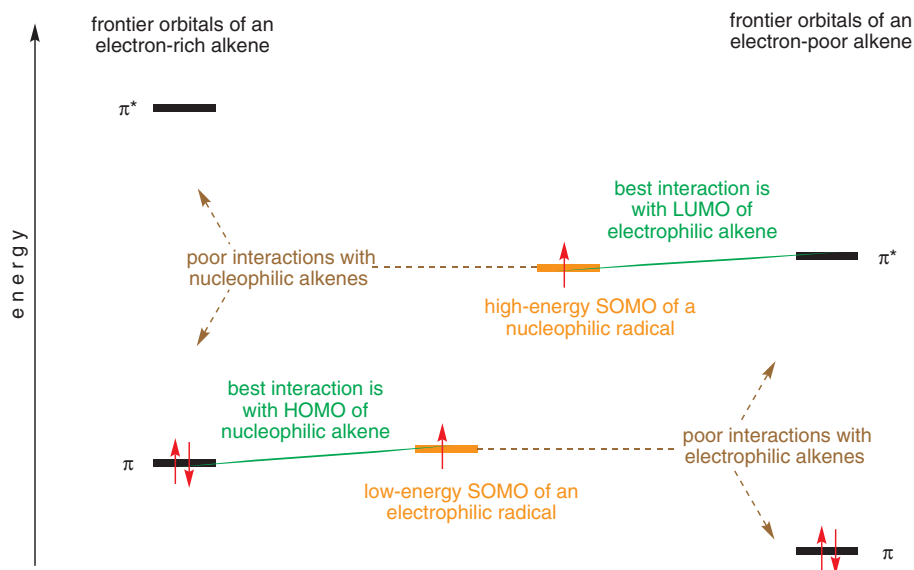
Alkene	% yield	Alkene	% yield
	95		85
	86		85
	72		83
	90		87

To explain why, we have to go back to our analysis (on p. 978) of the electronic structure of radicals and the energy of SOMOs. We said there that, while both electron-withdrawing groups and electron-donating groups will stabilize radicals, electron-withdrawing groups tend to lower the energy of the SOMO, while electron-donating groups tend to raise the energy of the SOMO.

● Electrophilic and nucleophilic radicals

- **Low-energy SOMOs** are more willing to accept an electron than to give one up; radicals adjacent to electron-withdrawing groups are therefore *electrophilic*.
- **High-energy SOMOs** are more willing to give up an electron than to accept an electron; radicals adjacent to electron-donating groups are therefore *nucleophilic*.

Hence the preferred reactivity of these alkyl radicals: they are relatively nucleophilic and therefore prefer to react with electrophilic alkenes. Reaction between a nucleophilic alkyl radical and an unfunctionalized (and therefore nucleophilic) alkene is much slower. Similarly, radicals adjacent to electron-withdrawing groups do not react well with electrophilic alkenes. We can represent all this on an energy level diagram.



We can now consider the third type of radical in the reaction mixture highlighted on p. 993—the **nitrile-stabilized alkyl radicals**. The diagram above explains the third aspect of radical chemoselectivity in the reaction: why both the product radical and the radicals produced by AIBN choose to react with Bu_3SnH and not with acrylonitrile. These radicals are electrophilic—they have an electron-withdrawing nitrile group attached to the radical centre, so reaction with an electron-poor alkene is slow.

Radical	Reacts with	Does not react with
$\text{R}-\dot{\text{C}}\text{H}-\text{CN}$	$\text{R}-\dot{\text{C}}\text{H}-\text{CN} + \text{H}-\text{SnBu}_3 \rightarrow \text{R}-\text{CH}_2-\text{CN} + \text{SnBu}_3\dot{\text{C}}\text{H}_2$	$\text{R}-\dot{\text{C}}\text{H}-\text{CN} + \text{CH}_2=\text{CH}-\text{CN} \rightarrow \text{R}-\text{CH}(\text{CN})-\dot{\text{C}}\text{H}-\text{CH}_2-\text{CN}$
$\text{NC}-\dot{\text{C}}(\text{R})_2$ [from AIBN]	$\text{NC}-\dot{\text{C}}(\text{R})_2 + \text{H}-\text{SnBu}_3 \rightarrow \text{NC}-\text{C}(\text{R})_2\text{H} + \text{SnBu}_3\dot{\text{C}}\text{H}_2$	$\text{NC}-\dot{\text{C}}(\text{R})_2 + \text{CH}_2=\text{CH}-\text{CN} \rightarrow \text{NC}-\text{C}(\text{R})_2-\dot{\text{C}}\text{H}-\text{CH}_2-\text{CN}$

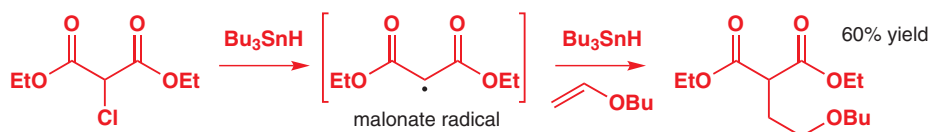
• Summary of requirements for the successful use of the tin method

Bu_3SnH	must be added or generated slowly
R-X starting material	must contain a weak C-X bond (C-I or C-Br)
radical trap	must be an electrophilic alkene must be present in a concentration at least 10 times that of Bu_3SnH

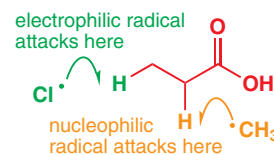
■ Notice that this reaction works even though a C-Cl bond needs to be broken to generate the radical. Usually only C-I and C-Br bonds can be used. However, this is a very weak C-Cl bond because the radical produced is so stable. This electrophilic radical can also be formed by H abstraction and by oxidation.

Electrophilic radicals

Having seen the energy diagram above, you will not be surprised to learn that the malonate radical adds readily not to electrophilic alkenes, but to nucleophilic alkenes, such as this vinyl ether, which carries an electron-donating oxygen substituent. The malonate radical is electron deficient; it has a low-energy SOMO which interacts best with the relatively high-energy HOMO of the electron-rich, nucleophilic alkene.



This difference in reactivity applies to non-carbon-centred radicals too. For example, the methyl radical $\cdot\text{CH}_3$ and the chlorine radical $\text{Cl}\cdot$ will both abstract a hydrogen atom from propionic acid. As you would expect, the methyl radical abstracts the hydrogen atom from next to the carbonyl group to form a carbonyl-stabilized radical. Perhaps surprisingly (in view of what we said earlier about the selectivity of radical chlorinations), the chlorine radical abstracts a hydrogen atom from the terminal methyl group of the acid, despite the fact that this C–H bond is stronger. The reason has to be to do with HOMO–LUMO interactions. The methyl radical is nucleophilic, with a high-energy SOMO. It therefore attacks the C–H bond with the lowest LUMO, in other words, α to the carbonyl group. The chlorine atom, on the other hand, is electrophilic: it has a low-energy SOMO (because it is an electronegative element) and attacks the C–H bonds of the terminal methyl group because they have the highest-energy HOMO. Chlorination of functionalized compounds is not as straightforward as that of simple alkanes!

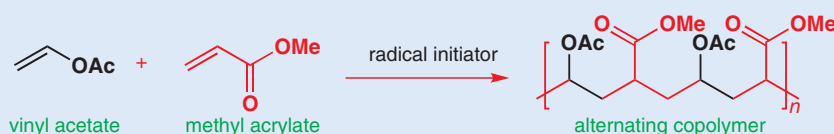


Interactive displays of common polymer structures

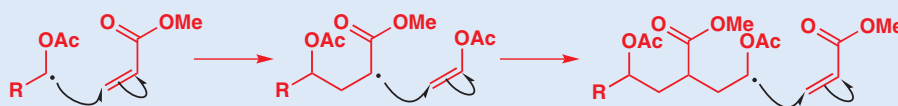
There is more on polymerization in the online chapter of that name

Copolymerization

Radical chain reactions are particularly suited to the synthesis of polymers, and there is one example of a polymerization that is worth including here since it demonstrates very nicely the effect of electron-withdrawing or -donating substituents on radical reactivity. When a mixture of vinyl acetate and methyl acrylate is treated with a radical initiator, a rather remarkable polymerization takes place. The polymer produced contains *alternating* vinyl acetate and methyl acrylate monomers along the length of its chain.



The mechanism of the reaction shows you why. The nucleophilic radical from vinyl acetate (adjacent to filled n orbital of OAc; high-energy SOMO) prefers to add to the electrophilic alkene (the acrylate). The new radical (adjacent to the empty π^* orbital of CO_2Me ; low-energy SOMO) is electrophilic and prefers to add to nucleophilic alkene (the vinyl acetate). This produces a new nucleophilic radical, which again prefers to add to the electrophilic alkene, and the whole cycle occurs repeatedly.



The radical produced by addition to vinyl acetate is nucleophilic, so it adds to methyl acrylate; the radical produced by addition to methyl acrylate is electrophilic, so it adds to vinyl acetate. This reaction is a clear demonstration of the power of frontier orbital theory to explain the reactivity of organic molecules—it would be hard to come up with any other convincing explanation.

The reactivity pattern of radicals is quite different from that of polar reagents

The first reaction that you met in this book, in Chapter 6, was nucleophilic addition to a carbonyl group. Yet we have shown you no examples of radicals adding to carbonyl groups. This typical reaction of polar reagents is really quite rare with radicals.

In Chapter 8 we introduced the concept of pK_a in which we saw acids and bases exchanging protons. Among the strongest organic acids are those containing O–H bonds. Yet you have seen no radical reactions in which an O–H bond is broken. Carbon acids tend to be much weaker—yet you’ve seen plenty of examples of C–H bonds being broken by radical attack.

In Chapter 15 we introduced nucleophilic substitution at saturated carbon, using as an example some alkyl bromides. Now, radicals do react with alkyl halides—but not at carbon! Instead they abstract the halogen, leaving an alkyl radical.

The difference in reactivity between, say, organolithiums and radicals, both of them highly reactive, is nicely illustrated by the way in which they react with enones.



➡ For the earlier discussions on hard and soft see p. 357 and 506.

➡ There is a striking illustration of the mismatch between the *reactivity* of bonds in a simple molecule and the strength of those bonds on p. 207.

We used the terms *hard* and *soft* in Chapters 15 and 22. From all these reactions it's evident that radicals are very soft species: their reactions are driven not at all by the charge density on an atom but by the strength of the bonds being attacked and by the coefficients and energies of the frontier orbitals. O–H bonds are easily broken by strong bases and C=O bonds attacked by strong nucleophiles because of the polarization in the O–H and C=O bond. O–H and C=O bonds are strong, and radicals care nothing for polarization, so radicals prefer to attack the much weaker C–H bonds which (because they are unpolarized) are often inert towards ionic reagents.

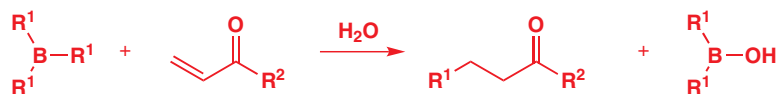
● Summary of typical reactivity patterns

With	Polar nucleophiles typically react like this	Radicals typically react like this
unsaturated C=O compounds		
X–H bonds		
alkyl halides		

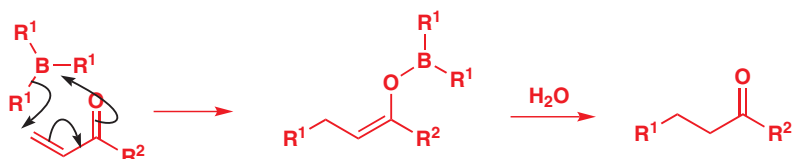
Alkyl radicals from boranes and oxygen

Although the tin hydride + alkyl halide method is very efficient, tin compounds are falling out of favour because of their toxicity. The reaction between boranes and oxygen provides a simple alternative, and many of the reactions carried out formerly using the tin hydride radical chemistry of pp. 993–997 can now be done using the method we are about to describe.

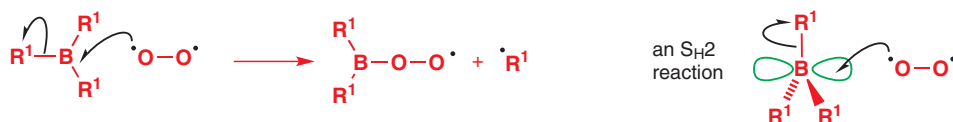
Mixing trialkylboranes with α,β -unsaturated carbonyl compounds in the presence of water gives conjugate addition of one of the alkyl groups from the boron. The carbonyl compound can be an aldehyde ($R^2 = H$) or a ketone ($R^2 = \text{alkyl}$).



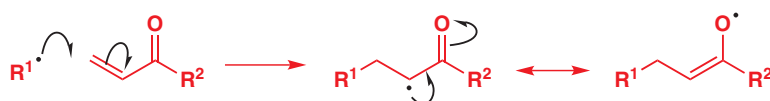
There was a dispute at first as to whether this was a radical reaction or an ionic reaction. The ionic reaction might have been a kind of pericyclic reaction with the intermediate boron enolate being hydrolysed by the water.



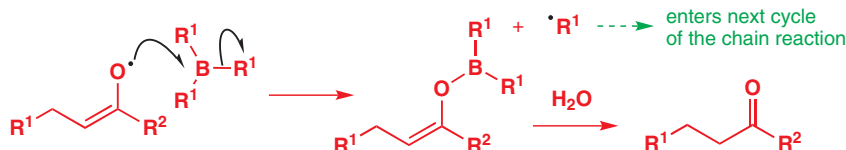
However, H. C. Brown discovered that the reaction was completely inhibited by just 5% of the stable radical galvinoxyl (shown on p. 975), known to be an efficient scavenger for radicals. But where were the radicals coming from? Further experiments showed that small amounts of oxygen were needed to make the reaction work. As you saw in Chapter 3, oxygen is a triplet diradical and displaces alkyl radicals from the trialkyl borane. This reaction looks at first like an S_N2 and is called an S_H2 (second order homolytic displacement), but in reality the oxygen adds to the empty p orbital of planar trigonal boron to release an alkyl radical and start the chain reaction.



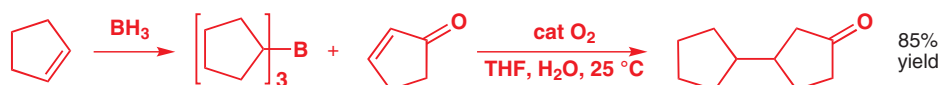
The alkyl radical now adds to the enone to give a delocalized intermediate that can be represented as a carbon- or oxygen-centred radical.



The chain is completed by displacement of another alkyl radical from a trialkylborane at oxygen by the delocalized radical and the formation of the same boron enolate proposed in the ionic reaction. This alkyl radical adds in its turn to the enone, and the boron enolate which forms is hydrolysed to the ketone product. Only small amounts of oxygen are needed to initiate the chain and it is not surprising that the air around the reaction mixture is enough to start a typical reaction. The water (which is inert to radicals, so can be present in the reaction mixture) again hydrolyses the boron enolate.

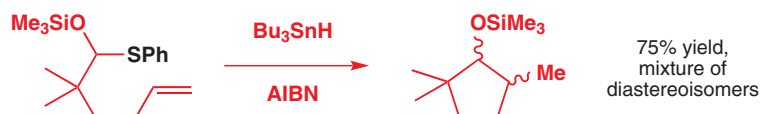


By combining a hydroboration step, which forms the borane, with the radical addition it is possible to carry out transformations such as the one below.



Intramolecular radical reactions are more efficient than intermolecular ones

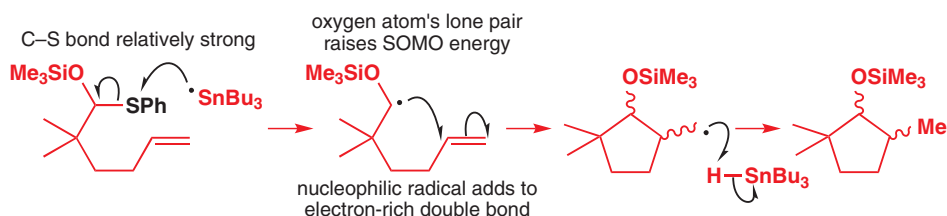
All of the reactions you have met so far involve radical attack of one molecule on another. We've pointed out some of the drawbacks when C–C bonds are made in this way: the radical trap has to be activated (that is, electrophilic to capture nucleophilic radicals) and must often be present in excess, and the radical starting material must contain very weak C–X bonds (such as C–Br, C–I). The requirements are much less stringent, however, if the radical reaction is intramolecular. For example, this reaction works:



➡ Explanation of the term 'triplet' in this context will be found in Chapter 38, p. 1010.

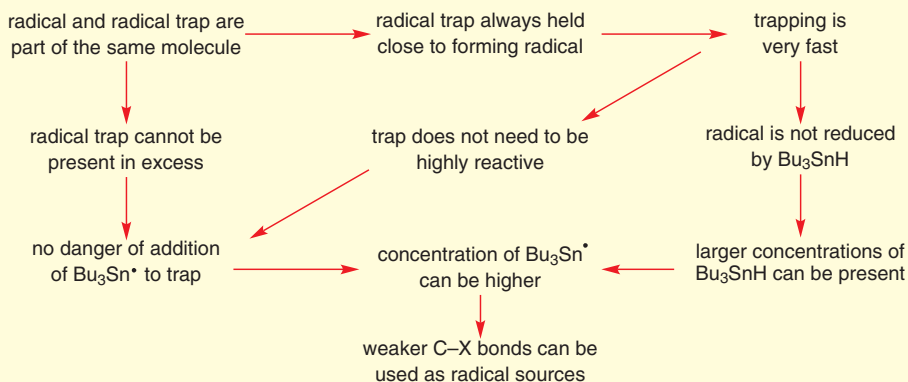
Notice that the double bond is not activated: in fact, it is nucleophilic, and the reaction still works even though the radical is also substituted with an electron-donating group. The C–S bond that is broken is also relatively strong, yet nonetheless a high yield of product is obtained. Why should this be so? What difference does it make that the reactions are intramolecular?

Interactive mechanism for intramolecular radical cyclization

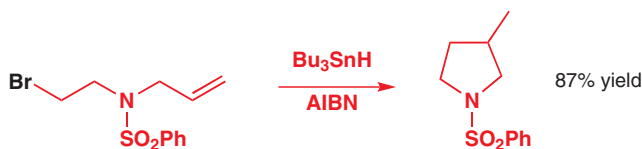


The key is that the intramolecular cyclization of the radical is now enormously favoured over other possible courses of action for the radical. Remember that when we were carrying out radical reactions *intermolecularly*, addition to the radical trap was encouraged by increasing the concentration of radical trap and decreasing the concentration of Bu_3SnH to avoid radical reduction. For *intramolecular* reactions, the double bond that acts as the radical trap is always held close to the radical, and cyclization takes place extremely rapidly, even on to unactivated double bonds. The hydride donor (Bu_3SnH) doesn't get a look in, and can be present in higher concentrations than would otherwise be possible. Moreover, as there is only one equivalent of radical trap, and the trap need not be highly reactive, there is little danger of high concentrations of $\text{Bu}_3\text{Sn}^\bullet$ reacting with it, so the concentration of $\text{Bu}_3\text{Sn}^\bullet$ can build up to levels where the rate of abstraction of groups like Cl, SPh, and SePh is acceptable, despite their stronger C–X bonds.

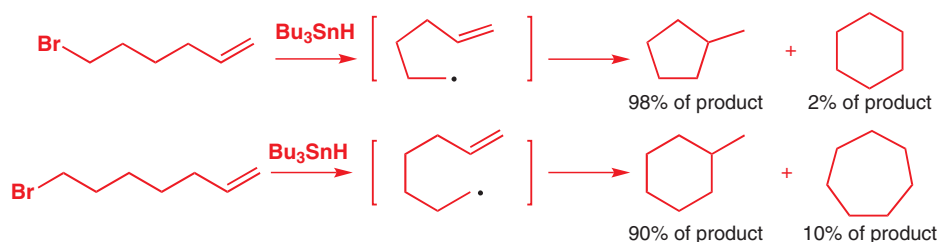
• Why are intramolecular radical reactions so good?



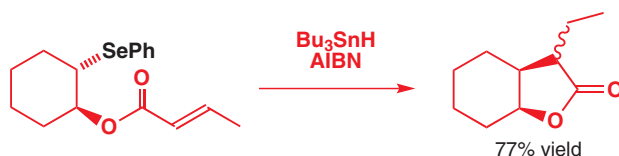
For all these reasons, intramolecular radical reactions are very powerful, and are often used to make five-membered rings.



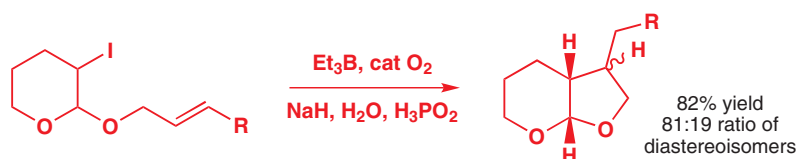
It is possible to make other ring sizes also, but the range is rather limited. Because of ring strain, three- and four-membered rings cannot be formed by radical reactions. Otherwise, smaller rings form faster than larger ones: look at these selectivities.



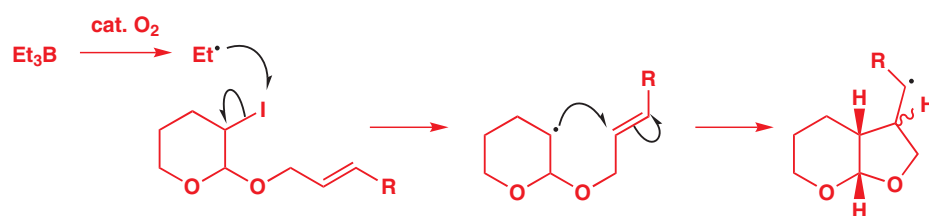
The preference for formation of a smaller ring is a very powerful one: in this reaction, the five-membered ring forms and not the six-membered one, even though cyclization to give a six-membered ring would also give a more stabilized radical.



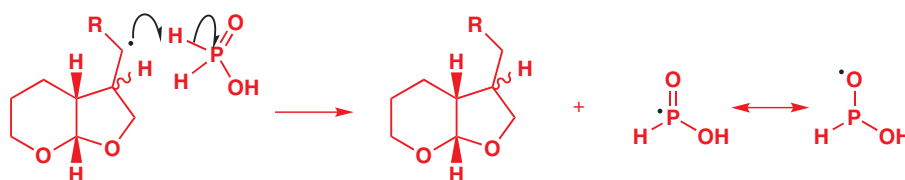
We said earlier that the toxicity of tin poses some problems, so it is useful that the borane-oxygen method (p. 998) works well for initiating radical cyclizations too. It is not necessary to incorporate boron into the starting material, since a combination of Et_3B , O_2 , and hypophosphorous acid, H_3PO_2 , can generate a radical from a halide which will cyclize in the same way as the tin-promoted examples you have just seen. Once again, a five-membered ring is preferred to the alternative six-membered ring.



Notice that the ethyl groups from Et_3B are not incorporated into the product. The displacement of $\text{Et}^\bullet$ from Et_3B initiates the chain reaction by abstracting the iodine atom from the starting material. The radical cyclizes to give a five-membered ring, as expected. A *cis* ring junction is inevitable because of the acetal 'tether'.



The product radical has to collect a hydrogen from somewhere, and this is the role of the hypophosphorous acid. Abstraction of H gives a radical that can be drawn either as P-centred or O-centred.

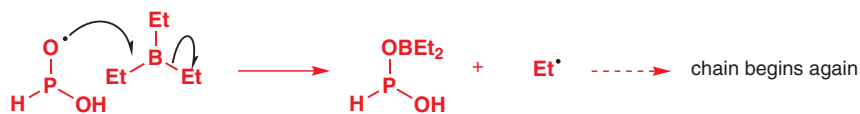


Interactive mechanism for radical ring closures

Baldwin's rules, describing the formation of different ring sizes, were described in Chapter 31, p. 810. They apply to radical reactions and ionic reactions equally well.

This type of stereochemical control is discussed in Chapter 32.

The chain is finally completed by a hydrogen abstraction from H_3PO_2 , which gives a radical that attacks the borane, just like oxygen did in the initiation step. A new ethyl radical is generated, which starts the cycle again.



Looking forward

Radicals are important because they react in ways difficult to achieve with anions and cations, and show usefully different selectivity. Although radical reactions are generally less important than ionic reactions, environmental and biological radical reactions are remarkably common in an atmosphere that is 20% oxygen diradical. Diradicals will feature to a greater extent in the next chapter, in which we move on from carbon atoms carrying seven valence electrons to carbon atoms carrying only six valence electrons, called *carbenes*.

Further reading

A basic introduction is *Radical Chemistry: The Fundamentals*, J. Perkins, Oxford Primer, OUP, Oxford, 2000. *Reactive Intermediates*, C. J. Moody and G. H. Whitham, Oxford Primer, OUP, Oxford, 2001, has a section on radicals.

The evidence that the McMurry reaction happens on a metal surface is quite nice, though, and if you're interested you can read McMurry's own account of it in *Accounts of Chemical Research*, 1983, **16**, 405 and J. E. McMurry, *Chem. Rev.*, 1989, **89**, 1513.

There are a couple of recipes in B. S. Furniss, A. J. Hannaford, P. W. G. Smith, and A. T. Tatchell, *Vogel's Textbook of Practical*

Organic Chemistry, Longman, 5th edn, 1989, pp. 576–579 that give examples of the use of NBS and dibenzoyl peroxide.

The borane-oxygen method of making radicals is reviewed by C. Ollivier and P. Renaud in *Chemical Reviews*, 2001, **101**, 3415. In common with most *Chemical Reviews*, this is a long scholarly article but reviews like this are essential to chemists wanting to know about a new reagent, method, or synthesis.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>